

Quantum Field Theory : *Renormalization* in a Nutshell

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Abstract

Renormalization is perhaps one of the most important discoveries of the last century physics. These notes aim to introduce the basic ideas of renormalization in the quickest possible way. We use the easiest example of the real scalar ϕ^4 theory to illustrate important concepts of renormalization. The article is self contained and assumes no prior knowledge of Quantum Field Theory.

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References

- *Mark Srednicki* : **Quantum Field Theory**
- *Diana Vaman* : **Modern Field Theory** (Lectures during Spring 2022 semester at the *University of Virginia*)
- *Micheal E. Peskin, Daniel V. Schroeder* : **An Introduction to Quantum Field Theory**
- *David Tong* : **Lecture Notes on Quantum Field Theory**
- *Sydney Coleman* : **Lecture Notes on Quantum Field Theory**

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1 Canonical Quantization

Historically, Canonical quantization is the method that was first used to quantize a field theory. In many ways, this technique is rather obsolete. But let us discuss it nonetheless.

Canonical quantization is the good old method that we used to “quantize” classical physics. In classical physics, a system is described by a lagrangian $L = L(q, \dot{q}, t)$. Given the lagrangian, we can define the *canonically conjugate momentum to the coordinate q* , which is denoted by p as,

$$p := \frac{\partial L}{\partial \dot{q}} \quad (1.1)$$

Then, we promote this classical theory to a quantum theory by imposing the following *canonical commutation relations*,

$$[q_a(t), q_b(t)] = 0 \quad (1.2)$$

$$[p_a(t), p_b(t)] = 0 \quad (1.3)$$

$$[q_a(t), p_b(t)] = i\delta_{ab} \quad (1.4)$$

We notice that the commutation relations are imposed at equal times. i.e, we do not have any commutation relations for $q(t)$ and $p(t')$ with $t' \neq t$. This is just a relic of the Heisenberg principle. Heisenberg is unhappy only if we measure the momentum and position of a particle at the *same time*!

Quantizing a field theory is no different. We write down a lagrangian, find the canonically conjugate momenta, and impose *equal time canonical commutation relations*. But there is one subtlety that has to be discussed. In quantum mechanics, position was an operator but time was just a parameter. If we want to study the quantum mechanics of relativistic systems, this is incompatible. So, we either have to promote time to an operator, or demote position to a mere parameter. In quantum field theory, the latter choice is made. Now, we consider the simplest possible field, the *real scalar field*.

1.1 The Real Scalar Field

Let $\phi = \phi(x)$ be a scalar field. Here, $x = x^\mu$ is a parameter. That is, we specify the value of the field at each space time point. We need to write down a lagrangian for a relativistic field theory of ϕ . This is known as the *Klein Gordon Lagrangian*, and is given by¹

$$\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi(x)\partial_\nu\phi(x) - \frac{1}{2}m^2\phi^2(x) \quad (1.5)$$

¹If you need some more arguments on how to “construct” this lagrangian, I recommend reading Sydney Coleman’s lecture notes.

This is the lagrangian for a *free* field theory. We are using the metric convention $\eta^{\mu\nu} = \text{diag}(-, +, +, +)$. If we vary the action with ϕ , we get the field equations, known as the *Klein Gordon equation*,

$$(-\square + m^2)\phi(x) = 0 \quad (1.6)$$

The field equation has linearly independent solutions given by, $\exp\{\pm ikx\}$, where we have used a short hand notation $kx = k^\mu x_\mu$. Plugging these into the equations of motion, we get,

$$(k^2 + m^2) \exp\{\pm ikx\} = 0 \quad (1.7)$$

So we must have,

$$k^2 = -m^2 \quad (1.8)$$

or,

$$(k^0)^2 = \vec{k}^2 + m^2 \quad (1.9)$$

This is the Einstein relation. So, in k^μ , the k^0 component is not independent. Therefore, we can construct a general solution for the field as,

$$\phi(x) = \int \frac{d^3k}{f(k)} \left(a(\vec{k}) e^{ikx} + b(\vec{k}) e^{-ikx} \right) \quad (1.10)$$

Since we are aiming at a Lorentz invariant theory, we choose $f(k)$ such that the measure is lorentz invariant. d^3k is not lorentz invariant because of the analogue of length contraction in momentum space. But d^4k is lorentz invariant because the γ factors due to lorentz transformation on dk^0 and d^3k cancel each other. We need to get rid of dk^0 integral from d^4k to get a lorentz invariant 3 measure. We know that k^0 is fixed by the Einstein relation, $k^2 = -m^2$. This is Lorentz invariant. Therefore, consider,

$$d^4k \delta(k^2 + m^2) \theta(k^0) \quad (1.11)$$

$\theta(k^0)$ is the Heavyside step function. This is included because we are only considering continuous lorentz transformations that do not reverse the time ordering. Now,

$$\begin{aligned} \delta(k^2 + m^2) &= \delta(-(k^0)^2 + \vec{k}^2 + m^2) \\ &= \frac{\delta(k^0 - \sqrt{\vec{k}^2 + m^2})}{2\sqrt{\vec{k}^2 + m^2}} + \frac{\delta(k^0 + \sqrt{\vec{k}^2 + m^2})}{2\sqrt{\vec{k}^2 + m^2}} \end{aligned} \quad (1.12)$$

Plugging this, the theta function gets rid of the second term, and we get,

$$d^4k \delta(k^2 + m^2) \theta(k^0) = d^3k dk^0 \frac{\delta(k^0 - \sqrt{\vec{k}^2 + m^2})}{2\sqrt{\vec{k}^2 + m^2}} \xrightarrow{\text{after } k^0 \text{ integral}} \frac{d^3k}{2\omega_{\vec{k}}} \quad (1.13)$$

where,

$$\omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2} \quad (1.14)$$

Choose,

$$f(k) = (2\pi)^3 \cdot 2\omega_{\vec{k}} \quad (1.15)$$

Denote,

$$\widetilde{dk} = \frac{d^3k}{(2\pi)^3 \cdot 2\omega_{\vec{k}}} \quad (1.16)$$

Then we have,

$$\phi(x) = \int \widetilde{dk} \left(a(\vec{k}) e^{ikx} + b(\vec{k}) e^{-ikx} \right) \quad (1.17)$$

Since we have a real scalar field, $\phi^* = \phi$. This gives us $b(\vec{k}) = a^*(\vec{k})$. Thus we have,

$$\phi(x) = \int \widetilde{dk} \left(a(\vec{k}) e^{ikx} + a^*(\vec{k}) e^{-ikx} \right) \quad (1.18)$$

Now, we find the canonical momentum. This is given by,

$$\Pi(x) := \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \quad (1.19)$$

Since,

$$\mathcal{L} = \frac{1}{2} \left(\dot{\phi}^2 - (\nabla \phi)^2 - m^2 \phi^2 \right) \quad (1.20)$$

we get,

$$\Pi(x) = \dot{\phi}(x) \quad (1.21)$$

We now quantize the theory. For that we put hats on top of dynamical variables and call them operators, and impose equal time commutation relations between coordinates and canonical momenta. So, we impose,

$$[\hat{\phi}(x), \hat{\phi}(y)]_{x^0=y^0} = 0 \quad (1.22)$$

$$[\hat{\Pi}(x), \hat{\Pi}(y)]_{x^0=y^0} = 0 \quad (1.23)$$

$$[\hat{\phi}(x), \hat{\Pi}(y)]_{x^0=y^0} = i\delta^3(\vec{x} - \vec{y}) \quad (1.24)$$

Now, using these along with the $a, a^\dagger$ expansion of the fields, we can verify²,

$$[\hat{a}(\vec{k}), \hat{a}(\vec{k}')] = 0 \quad (1.25)$$

$$[\hat{a}^\dagger(\vec{k}), \hat{a}^\dagger(\vec{k}')] = 0 \quad (1.26)$$

$$[\hat{a}(\vec{k}), \hat{a}^\dagger(\vec{k}')] = (2\pi)^3 \cdot 2\omega_{\vec{k}} \delta^3(\vec{k} - \vec{k}') \quad (1.27)$$

²Of course, it is left as an exercise for the reader!

1.2 The Spectrum of the Theory

To find the spectrum of the theory, we first find the Hamiltonian. The Hamiltonian is,

$$H = \int d^3x \mathcal{H} \quad (1.28)$$

where,

$$\mathcal{H} = \Pi \dot{\phi} - \mathcal{L} \quad (1.29)$$

This is a trivial thing to find, and we get,

$$\mathcal{H} = \frac{\Pi^2(x)}{2} + \frac{(\nabla \phi(x))^2}{2} + \frac{1}{2} m^2 \phi^2(x) - \Omega_0 \quad (1.30)$$

Notice that we have added an extra factor Ω_0 here. We can think of it as a constant added to the lagrangian density, which does not change the field equations at all. This is added for a later convenience. Now, we get the Hamiltonian to be,

$$H = -\Omega_0 V + \int d^3x \left(\frac{\Pi^2(x)}{2} + \frac{(\nabla \phi(x))^2}{2} + \frac{1}{2} m^2 \phi^2(x) \right) \quad (1.31)$$

Now,

$$\begin{aligned} \int d^3x \frac{\Pi^2(x)}{2} &= \frac{1}{2} \int d^3x \widetilde{dk} \widetilde{dp} \left\{ \left[-i\omega_{\vec{k}} a(\vec{k}) e^{ikx} + i\omega_{\vec{k}} a^\dagger(\vec{k}) e^{-ipx} \right] \left[-i\omega_{\vec{p}} a(\vec{p}) e^{ipx} + i\omega_{\vec{p}} a^\dagger(\vec{p}) e^{-ipx} \right] \right\} \\ &= \frac{1}{2} \int \widetilde{dk} \widetilde{dp} \left\{ (2\pi)^3 \delta^3(\vec{k} - \vec{p}) \omega_{\vec{k}} \omega_{\vec{p}} \left[a(\vec{k}) a^\dagger(\vec{p}) e^{i(k^0 - p^0)t} + a^\dagger(\vec{k}) a(\vec{p}) e^{-i(k^0 - p^0)t} \right] \right. \\ &\quad \left. - (2\pi)^3 \delta^3(\vec{k} + \vec{p}) \omega_{\vec{k}} \omega_{\vec{p}} \left[a(\vec{k}) a(\vec{p}) e^{i(k^0 + p^0)t} + a^\dagger(\vec{k}) a^\dagger(\vec{p}) e^{-i(k^0 + p^0)t} \right] \right\} \\ &= \frac{1}{2} \int \widetilde{dk} \frac{1}{2\omega_{\vec{k}}} \left[(\omega_{\vec{k}}^2) \left(a(\vec{k}) a^\dagger(\vec{k}) + a^\dagger(\vec{k}) a(\vec{k}) \right) + (-\omega_{\vec{k}}^2) (\text{something}) \right] \end{aligned} \quad (1.32)$$

Similarly finding other two terms, we end up with,

$$H = -\Omega_0 V + \frac{1}{2} \int \widetilde{dk} \frac{1}{2\omega_{\vec{k}}} \left[\left(\omega_{\vec{k}}^2 + \vec{k}^2 + m^2 \right) \left(a(\vec{k}) a^\dagger(\vec{k}) + a^\dagger(\vec{k}) a(\vec{k}) \right) + \left(-\omega_{\vec{k}}^2 + \vec{k}^2 + m^2 \right) (\text{something}) \right] \quad (1.33)$$

Using Einstein relation, we get,

$$H = -\Omega_0 V + \int \widetilde{dk} \frac{\omega_{\vec{k}}}{2} \left(a(\vec{k}) a^\dagger(\vec{k}) + a^\dagger(\vec{k}) a(\vec{k}) \right) \quad (1.34)$$

Now, the commutation relations gives us,

$$a(\vec{k}) a^\dagger(\vec{k}) = a^\dagger(\vec{k}) a(\vec{k}) + (2\pi)^3 \cdot 2\omega_{\vec{k}} \delta^3(\vec{0}) \quad (1.35)$$

Plugging this in,

$$H = \int \widetilde{dk} \omega_{\vec{k}} a^\dagger(\vec{k}) a(\vec{k}) + \int d^3k \frac{\omega_{\vec{k}}}{2} \delta^3(\vec{0}) - \Omega_0 V \quad (1.36)$$

The second term here is infinite! There are two infinities hidden in the term. The first one is,

$$\delta^3(\vec{0}) = \int d^3x e^{i\vec{k}\cdot\vec{x}} \Big|_{\vec{k}=0} = \int d^3x = V \quad (1.37)$$

This infinity is coming from the fact that space is infinite. We remedy by this by the fact that we cannot measure the energy of all space, we can only measure energy densities. The other part of the infinity is,

$$\int d^3k \frac{\omega_{\vec{k}}}{2} \quad (1.38)$$

For large values of $\vec{k}$, $\omega \sim |\vec{k}|$, then,

$$\int d^3k \frac{\omega_{\vec{k}}}{2} \sim \int_0^\infty dk k^2 \times k \longrightarrow \text{diverge quartically} \quad (1.39)$$

This divergence arises from the fact that we assumed the theory to be valid for all momenta/energy. This is of course not the case! These divergences due to large momenta plague quantum field theory a lot. Now, we fix it the following way: let the largest value of momenta possible be some large Λ . Then,

$$\int d^3k \frac{\omega_{\vec{k}}}{2} = \varepsilon_0 \quad (1.40)$$

Hence,

$$H = \int \widetilde{dk} \omega_{\vec{k}} a^\dagger(\vec{k}) a(\vec{k}) + (\varepsilon_0 - \Omega_0) V \quad (1.41)$$

We choose $\Omega_0 = \varepsilon_0$. This is our first glimps at renormalization! Therefore, we have,

$$H = \int \widetilde{dk} \omega_{\vec{k}} a^\dagger(\vec{k}) a(\vec{k}) \quad (1.42)$$

States

Let $|E\rangle$ be an energy eigen state. Then we can show that³ $a^\dagger(\vec{k})|E\rangle$ is also an eigen state with energy $E + \omega_{\vec{k}}$ and $a(\vec{k})|E\rangle$ is an eigen state with energy $E - \omega_{\vec{k}}$. We claim that $E \geq 0$. This is because, $H \sim a^\dagger a$ and $E = \langle E|H|E\rangle = \langle E|a^\dagger a|E\rangle$. Now if we define $|A\rangle = a|E\rangle$, then $E = \langle A|A\rangle \geq 0$. This along with the fact that a lowers the energy tells us that there

³This is the same procedure as the Harmonic oscillator algebra!

must be a state which is annihilated by a . Let us denote it by $|0\rangle$ and call it the vacuum. We have,

$$a(\vec{\mathbf{k}}) |0\rangle = 0 \quad (1.43)$$

Therefore,

$$H |0\rangle = \int \widetilde{dk} \omega_{\vec{\mathbf{k}}} a^\dagger(\vec{\mathbf{k}}) a(\vec{\mathbf{k}}) |0\rangle = 0 \quad (1.44)$$

The vacuum has zero energy. We can also show that it has zero momentum. Now if we define,

$$|p\rangle = a^\dagger(\vec{\mathbf{p}}) |0\rangle \quad (1.45)$$

Then,

$$\begin{aligned} H |p\rangle &= \int \widetilde{dk} \omega_{\vec{\mathbf{k}}} a^\dagger(\vec{\mathbf{k}}) a(\vec{\mathbf{k}}) a^\dagger(\vec{\mathbf{p}}) |0\rangle \\ &= \int \widetilde{dk} \omega_{\vec{\mathbf{k}}} a^\dagger(\vec{\mathbf{k}}) \left[a^\dagger(\vec{\mathbf{p}}) a(\vec{\mathbf{k}}) + (2\pi)^3 \cdot 2\omega_{\vec{\mathbf{k}}} \delta^3(\vec{\mathbf{k}} - \vec{\mathbf{p}}) \right] |0\rangle \\ &= 0 + \omega_{\vec{\mathbf{p}}} a^\dagger(\vec{\mathbf{p}}) |0\rangle \\ &= \omega_{\vec{\mathbf{p}}} |p\rangle \end{aligned} \quad (1.46)$$

Hence $|p\rangle$ is a state with energy $\omega_{\vec{\mathbf{p}}}$. We interpret this as a particle. Therefore in quantum field theory, a one particle state is the first excited state of the Hamiltonian.

2 Scattering Amplitudes : LSZ Formula

One of the most important kind of experiments are the scattering experiment. To study the properties of something, we smash particles into it and see what comes out. Here, we smash particles into each other to study the particles themselves. The basic idea of a scattering process is this: we send two (or more) essentially non interacting particles at each other from time $-\infty$, the particles interact for a while and we observe essentially non interacting particles at time $+\infty$. These asymptotic times of observations are very large compared to the time scales of interaction, and hence we take them to be infinite. Let us try to formulate this process in Quantum Field Theory.

2.1 Review of Free Theory

For a free Real Scalar Field theory, we had,

$$\phi_0(x) = \int \widetilde{dp} \left(a_0(\vec{p}) e^{ip \cdot x} + a_0^\dagger(\vec{p}) e^{-ip \cdot x} \right) \quad (2.1)$$

where,

$$\widetilde{dp} = \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_{\vec{p}}} \quad (2.2)$$

and

$$p^0 = \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2} \quad (2.3)$$

a_0 and $a_0^\dagger$ are the creation and annihilation operators in the free theory. The subscript 0 is added to denote that these are free field operators. Let $|0\rangle$ be the vacuum of the free theory, with zero energy and momentum. The one particle state is defined by,

$$|p\rangle \equiv a_0^\dagger(\vec{p}) |0\rangle \quad (2.4)$$

The normalisation of the one particle state is,

$$\langle p|q\rangle = (2\pi)^3 2\omega_{\vec{p}} \delta^3(\vec{p} - \vec{q}) \quad (2.5)$$

We have,

$$\phi_0(x) |0\rangle = \int \widetilde{dp} e^{-ip \cdot x} |p\rangle \quad (2.6)$$

Then the normalisation condition can be rewritten as,

$$\langle p| \phi_0(x) |0\rangle = e^{-ip \cdot x} \quad (2.7)$$

We would like to create wave packets. Let $f_1(\vec{k})$ be a wave packet localised in the momentum space around $\vec{k}_1$. i.e,

$$f_1(\vec{k}) \propto \exp\left\{-\frac{(\vec{k} - \vec{k}_1)^2}{4\sigma^2}\right\} \quad (2.8)$$

Then we can define,

$$(a_1^\dagger)_0 = \int \frac{d^3k}{(2\pi)^3} f_1(\vec{k}) a_0^\dagger(\vec{k}) \quad (2.9)$$

This will create a wave packet centered at $\vec{k}_1$ in the momentum space. Using,

$$a_0^\dagger(\vec{p}) = -i \int d^3x e^{ip \cdot x} \overleftrightarrow{\partial}_0 \phi_0(x) \quad (2.10)$$

Where $p^0 = \omega_{\vec{p}}$. We can write,

$$(a_1^\dagger)_0 = -i \int \frac{d^3k}{(2\pi)^3} f_1(\vec{k}) \int d^3x e^{ip \cdot x} \overleftrightarrow{\partial}_0 \phi_0(x) \quad (2.11)$$

We can use this definition to create particle wave packets which are separated widely at asymptotic times. We do the following. Let $\vec{k}_1 \neq \vec{k}_2$. Then $(a_1^\dagger)_0(a_2^\dagger)_0|0\rangle$ is a two particle state, with one particle having momentum $\vec{k}_1$ and the other $\vec{k}_2$. Since their momenta are different, if we allow the states to evolve in time in the Schrodinger picture, they are guaranteed to be widely separated in the far future and remote past.

2.2 Assumptions of LSZ

We would like to do the exact same thing in an interacting field theory. That is to create essentially free particles in the infinite past, throw them at each other and see what comes out at the far future. Let us denote the fully interacting field by $\phi(x)$. It is now not very clear about how to write the field in terms of analogous creation and annihilation operators.

We notice that if we freeze time, i.e, fix $t = t_0$, then we can use the free field expression for $\phi_0(x)$ at that instant of time for $\phi(t_0, \vec{x})$. We can understand this physically the following way; at one instant of time, the field at each point has some value. Then at each point (in momentum space), we can write the field to be a linear combination of harmonic oscillator solutions⁴. This is exactly what the free field expression is doing. At each point in the momentum space, expand the field in Harmonic oscillator basis, then take the sum over all possible momenta. Therefore, for a fixed time $t = t_0$, we have⁵,

⁴Harmonic oscillator solutions form a complete basis

⁵Here, a_0 do not have the interpretation of annihilation operator of one particle. The interpretation is not really clear to me.

$$\phi(t_0, \vec{x}) = \int \widetilde{d\vec{p}} \left(a_0(\vec{p}) e^{ip \cdot x} + a_0^\dagger(\vec{p}) e^{-ip \cdot x} \right) \quad (2.12)$$

with $p^0 = \omega_{\vec{p}}$ and $x^0 = t_0$. Now, how do we get the expression for $\phi(x)$ for any time t ? We evolve $\phi(t_0, \vec{x})$ using the full Hamiltonian of the theory⁶. That is,

$$\phi(t, \vec{x}) \equiv \phi(x) = e^{iH(t-t_0)} \phi(t_0, \vec{x}) e^{-iH(t-t_0)} \quad (2.13)$$

with,

$$H = H_0 + H_1 \quad (2.14)$$

where

$$H_1 = \int d^3x \frac{\lambda}{4!} \phi^4(x) \quad (2.15)$$

The time evolution on $\phi(t_0, \vec{x})$, changes a_0 and $a_0^\dagger$. It changes say $a_0^\dagger$ to all possible combinations of $(a_0^\dagger)^n$ for all n . This can be sort of inferred from the Baker Campbell formula. All these arguments were sort of hand waving. So, we use these as a motivation for the following assumption,

The interacting field can be expanded the following way,

$$\phi(x) = \sum_{\lambda} \int \widetilde{d\vec{k}} \left(a(t, \vec{k}; \lambda) e^{ik \cdot x} + a^\dagger(t, \vec{k}; \lambda) e^{-ik \cdot x} \right) \quad (2.16)$$

with $k^0 = \omega_{\vec{k}}$, and $a^\dagger(t, \vec{k}; \lambda)$ creates a state with λ number of particles.

What good does it do for us? Our goal is to prove that it is possible to create correctly normalized one particle states at asymptotic times ($t \rightarrow \pm\infty$). If we are able to prove this, then we are ready for scattering experiments. Create $|i\rangle$, a free particle state at $t \rightarrow -\infty$, evolve it in time, then look at the overlap with $|f\rangle$ at $t \rightarrow \infty$ which is also a collection of one particle states. We are interested in the scattering matrix given by,

$$S_{fi} = \langle f; +\infty | i; -\infty \rangle \quad (2.17)$$

In free theory, $S_{fi} = \delta_{fi}$.

Now we proceed to prove that it is possible to create correctly normalized one particle states at asymptotic times. Let $|\Omega\rangle$ be the vacuum of the interacting theory. We can always choose it to have zero energy and momentum. We need $\phi(x)$ to act on the vacuum and not produce a state with an overlap with the vacuum itself. This is precisely because we need

⁶Remember ϕ is an operator!

$a^\dagger(\pm\infty)|\Omega\rangle$ to be a one particle state, and should not have an overlap with the vacuum. Now,

$$\langle\Omega|\phi(x)|\Omega\rangle = \langle\Omega|e^{-iP_\mu x^\mu}\phi(0)e^{iP_\mu x^\mu}|\Omega\rangle = \langle\Omega|\phi(0)|\Omega\rangle \quad (2.18)$$

The last equality follows because $P_\mu|\Omega\rangle = 0$. The last quantity is a Lorentz invariant number⁷.

We need the above matrix element to be zero. We can always make it zero by shifting the field $\phi(x) \rightarrow \phi(x) + v$ where $v = \langle\Omega|\phi(0)|\Omega\rangle$. This is not going to change any physics at all. So, without any generality loss, we set the above to zero.

Now, let us denote a one particle state in the interacting theory by $|1_{\vec{k}}\rangle$, two particle state by $|2_{\vec{k}}, n\rangle$ and so on. The n in multi particle states denote the additional parameters the state have, like the relative momenta etc. The subscript $\vec{k}$ in the multiparticle state is the total momentum of the state. If $|\lambda_{\vec{k}}, n\rangle$ is a multiparticle state, then it is associated with an energy momentum dispersion relation given by,

$$\omega_{\vec{k}}(\lambda) = \sqrt{\vec{k}^2 + m_\lambda^2} \quad (2.19)$$

Here, m_λ is the analogous mass term. In the case of one particle state, $m_\lambda = m$. In other cases, $m_\lambda = M \geq 2m$.

Now consider,

$$\langle 1_{\vec{k}}|\phi(x)|\Omega\rangle = \langle 1_{\vec{k}}|e^{-iP_\mu x^\mu}\phi(0)e^{iP_\mu x^\mu}|\Omega\rangle = e^{-ik\cdot x}\langle 1_{\vec{k}}|\phi(0)|\Omega\rangle \quad (2.20)$$

Since we can get to $|1_{\vec{k}}\rangle$ by a Lorentz boost on $|1_{\vec{0}}\rangle$, and $\phi(0)$ is Lorentz invariant, we can write,

$$\langle 1_{\vec{k}}|\phi(x)|\Omega\rangle = e^{-ik\cdot x}\langle 1_{\vec{0}}|\phi(0)|\Omega\rangle = e^{-ik\cdot x}\sqrt{Z} \quad (2.21)$$

For $\phi(x)$ to create a correctly normalized state⁸ at asymptotic times, we need $Z = 1$. This can be achieved by simply re-scaling $\phi(x) \rightarrow \sqrt{Z}\phi(x)$. This is called the *Field Strength Renormalization*. We assume this process is done.

Next, we must make sure that $\phi(x)$ do not create any multi particle states at asymptotic times. For that, it is enough to make sure that $\langle\lambda_{\vec{k}}, n|a_1^\dagger(\pm\infty)|\Omega\rangle = 0$, where $\lambda \neq 1$. To do this, we need an expression for $a_1^\dagger(t)$, which is the interacting field analogue for the normalized creation operator. For that, we can just invert equation (2.16) and find⁹,

$$a^\dagger(t, \vec{k}) = \sum_\lambda a^\dagger(t, \vec{k}; \lambda) = -i \int d^3x e^{ik\cdot x} \overleftrightarrow{\partial}_0 \phi(x) \quad (2.22)$$

⁷This is because 0 do not transform under Lorentz transformations

⁸correct normalization is the normalization for free theory

⁹Deriving this for the free theory can be found in chapter 3 of Srednicki

Here, $f \overset{\leftrightarrow}{\partial} g := f \partial g - g \partial f$.

Then we can define a normalized creation operator,

$$a_1^\dagger(t) = -i \sum_{\lambda} \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{k}) \int d^3 x e^{ik \cdot x} \overset{\leftrightarrow}{\partial}_0 \phi(x) \quad (2.23)$$

Now armed with the above expansion, we can find $\langle \lambda_{\vec{p}}, n | a_1^\dagger(\pm\infty) | \Omega \rangle$. But since $a_1^\dagger(t)$ creates a normalisable state, we should compare it with another normalisable state. Define,

$$|\psi_{\lambda'}\rangle = \int \frac{d^3 p}{(2\pi)^3} \psi_n(\vec{p}) |\lambda'_{\vec{p}}, n\rangle \quad (2.24)$$

Where $\psi_n(\vec{p})$ is an appropriate normalizable wave packet. The summation over the collective index n is omitted. Now,

$$\langle \psi_{\lambda'} | a_1^\dagger(t) | \Omega \rangle = -i \int \frac{d^3 p}{(2\pi)^3} \psi_n^*(\vec{p}) \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{k}) \sum_{\lambda} \int d^3 x e^{ik \cdot x} \overset{\leftrightarrow}{\partial}_0 \langle \lambda'_{\vec{p}}, n | \phi(x) | \Omega \rangle \quad (2.25)$$

But

$$\langle \lambda'_{\vec{p}}, n | \phi(x) | \Omega \rangle = e^{-ip \cdot x} \langle \lambda'_{\vec{0}}, n | \phi(0) | \Omega \rangle = e^{-ip \cdot x} A_n(\lambda', \vec{p}) \quad (2.26)$$

$A_n(\lambda', \vec{p})$ is a Lorentz invariant function. Also, $p^0 = \omega_{\vec{p}}(\lambda')$. Then we get,

$$\begin{aligned} \langle \psi_{\lambda'} | a_1^\dagger(t) | \Omega \rangle &= -i \int \frac{d^3 p}{(2\pi)^3} \psi_n^*(\vec{p}) \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{k}) \sum_{\lambda} \int d^3 x e^{ik \cdot x} \overset{\leftrightarrow}{\partial}_0 e^{-ip \cdot x} A_n(\lambda', \vec{p}) \\ &= - \int \frac{d^3 p}{(2\pi)^3} \psi_n^*(\vec{p}) \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{k}) \sum_{\lambda} \int d^3 x e^{i(k-p) \cdot x} (p^0 + k^0) A_n(\lambda', \vec{p}) \\ &= - \int \frac{d^3 p}{(2\pi)^3} \psi_n^*(\vec{p}) \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{k}) \sum_{\lambda} e^{i(p^0 - k^0)t} (2\pi)^3 \delta^3(\vec{p} - \vec{k}) (p^0 + k^0) A_n(\lambda', \vec{p}) \\ &= - \sum_{\lambda} \int \frac{d^3 p}{(2\pi)^3} \psi_n^*(\vec{p}) f_1(\vec{p}) (p^0 + k^0) A_n(\lambda', \vec{p}) e^{i(p^0 - k^0)t} \end{aligned}$$

Notice that $k^0 = \omega_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$ and $p^0 = \omega_{\vec{p}}(\lambda') = \sqrt{\vec{p}^2 + m_{\lambda'}^2} > k^0$, and the inequality is strict. Therefore if we take $t \rightarrow \pm\infty$, then due to the wildly oscillating factor, the above expression will go to zero!

Therefore, if we take the field expansion in (2.16), then we are guaranteed that the field will create correctly normalized one particle states out of the vacuum after a shift and re-scaling of the field!

2.3 Derivation of the Formula

Now, we go on to derive the *LSZ Reduction Formula*. We can now create asymptotically free states in the interacting theory. Let us create a two particle state,

$$|i\rangle = \lim_{t \rightarrow -\infty} a_1^\dagger(t) a_2^\dagger(t) |\Omega\rangle \quad (2.27)$$

The particles asymptotically will have momenta centered around $\vec{\mathbf{k}}_1$ and $\vec{\mathbf{k}}_2$ respectively. In the far future, we look for states having momenta $\vec{\mathbf{k}}'_1$ and $\vec{\mathbf{k}}'_2$, i.e.,

$$|f\rangle = \lim_{t \rightarrow \infty} a_{1'}^\dagger(t) a_{2'}^\dagger(t) |\Omega\rangle \quad (2.28)$$

We need,

$$S_{fi} = \langle f|i\rangle = \langle \Omega| a_{1'}(\infty) a_{2'}(\infty) a_1^\dagger(-\infty) a_2^\dagger(-\infty) |\Omega\rangle \quad (2.29)$$

Since the operators are time ordered already, we can insert the time ordering operator without changing anything. then,

$$\langle f|i\rangle = \langle \Omega| \mathcal{T} a_{1'}(\infty) a_{2'}(\infty) a_1^\dagger(-\infty) a_2^\dagger(-\infty) |\Omega\rangle \quad (2.30)$$

It would be nice if we could move the “daggered” guys to the left. For that, we do the following,

$$\begin{aligned} a_1^\dagger(\infty) - a_1^\dagger(-\infty) &= \int_{-\infty}^{\infty} dt \frac{\partial a_1^\dagger(t)}{\partial t} \\ &= -i \sum_{\lambda} \int_{-\infty}^{\infty} dt \partial_0 \left[\int \frac{d^3 k}{(2\pi)^3} f_1(\vec{\mathbf{k}}) \int d^3 x e^{ik \cdot x} \overset{\leftrightarrow}{\partial}_0 \phi(x) \right] \\ &= -i \sum_{\lambda} \int d^4 x \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{\mathbf{k}}) \partial_0 \left[e^{ik \cdot x} \overset{\leftrightarrow}{\partial}_0 \phi(x) \right] \\ &= -i \sum_{\lambda} \int d^4 x \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{\mathbf{k}}) \left[e^{ik \cdot x} \partial_0^2 \phi(x) - (\partial_0^2 e^{ik \cdot x}) \phi(x) \right] \end{aligned} \quad (2.31)$$

But we can use,

$$(\partial_0^2 - \nabla^2 + m^2) e^{ik \cdot x} = 0 \quad (2.32)$$

Also, we can drop the sum over λ because we have already shown that at asymptotic times, only $\lambda = 1$ term is non zero. Then,

$$a_1^\dagger(\infty) - a_1^\dagger(-\infty) = -i \int d^4 x \int \frac{d^3 k}{(2\pi)^3} f_1(\vec{\mathbf{k}}) \left[e^{ik \cdot x} \partial_0^2 \phi(x) + ((-\nabla^2 + m^2) e^{ik \cdot x}) \phi(x) \right] \quad (2.33)$$

Using integration by parts twice, we can convert the ∇^2 onto $\phi(x)$. Then,

$$\begin{aligned} a_1^\dagger(\infty) - a_1^\dagger(-\infty) &= -i \int d^4x \int \frac{d^3k}{(2\pi)^3} f_1(\vec{k}) [e^{ik \cdot x} (\partial_0^2 - \nabla^2 + m^2) \phi(x)] \\ &= -i \int d^4x \int \frac{d^3k}{(2\pi)^3} f_1(\vec{k}) e^{ik \cdot x} (-\square + m^2) \phi(x) \end{aligned} \quad (2.34)$$

Hence, we can write,

$$a_1^\dagger(-\infty) = a_1^\dagger(\infty) + i \int d^4x \int \frac{d^3k}{(2\pi)^3} f_1(\vec{k}) e^{ik \cdot x} (-\square + m^2) \phi(x) \quad (2.35)$$

and

$$a_1(\infty) = a_1(-\infty) + i \int d^4x \int \frac{d^3k}{(2\pi)^3} f_1(\vec{k}) e^{-ik \cdot x} (-\square + m^2) \phi(x) \quad (2.36)$$

We can now use these in the calculation of S_{fi} . Then all the “daggers” come to the left and kill the vacuum, and “undaggered” guys go to the right to kill the vacuum¹⁰. We also take the limit where $f_1(\vec{k}) \rightarrow (2\pi)^3 \delta^3(\vec{k} - \vec{k}_1)$. Then we have,

$$\begin{aligned} \langle f|i \rangle &= (i)^{2+2} \int d^4x_1 d^4x_2 d^4x_{1'} d^4x_{2'} e^{ik_1 \cdot x_1} e^{ik_2 \cdot x_2} e^{-ik_{1'} \cdot x_{1'}} e^{-ik_{2'} \cdot x_{2'}} \\ &\quad \times (-\square_1^2 + m^2) (-\square_2 + m^2) (-\square_{1'} + m^2) (-\square_{2'} + m^2) \\ &\quad \times \langle \Omega | \mathcal{T} \phi(x_1) \phi(x_2) \phi(x_{1'}) \phi(x_{2'}) | \Omega \rangle \end{aligned} \quad (2.37)$$

In general for n particle to l particle scattering,

$$\begin{aligned} \langle \vec{k}'_1 \cdots \vec{k}'_l | S | \vec{k}_1 \cdots \vec{k}_n \rangle &= (i)^{n+l} \int \prod_{i=1}^n d^4x_i e^{ik_i \cdot x_i} (-\square_{x_i} + m^2) \prod_{j=1}^l d^4y_j e^{-ik'_j \cdot y_j} (-\square_{y_j} + m^2) \\ &\quad \times \langle \Omega | \mathcal{T} \phi(x_1) \cdots \phi(x_n) \phi(y_1) \cdots \phi(y_l) | \Omega \rangle \end{aligned} \quad (2.38)$$

L.H.S. is written in the Heisenberg picture. This is the *LSZ Reduction Formula!*

2.4 Implications of Re-scaling and Shifting

We had our ϕ^4 theory lagrangian to be,

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m_0^2 \phi^2 - \frac{\lambda_0}{4!} \phi^4 \quad (2.39)$$

¹⁰Thanks to the time ordering symbol, we don't have to worry about commutators or anything!

Here, we need to do the shifting and re-scaling the field. The shifting is done to make sure that $\langle \Omega | \phi(x) | \Omega \rangle$ is zero. Since the original lagrangian has $\phi \rightarrow -\phi$ symmetry, this term is already zero! Therefore, we only need to re-scale the field. Upon re-scaling, we get,

$$\mathcal{L} = -\frac{1}{2}Z\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}Zm_0^2\phi^2 - \frac{\lambda_0}{4!}Z^2\phi^4 \quad (2.40)$$

Define,

$$Zm_0^2 = Z_m m^2 \quad (2.41)$$

and

$$\lambda_0 Z^2 = Z_\lambda \lambda \quad (2.42)$$

Where, m and λ are the physical mass and coupling constants that we measured from experiments, whereas m_0 and λ_0 are just parameters of the lagrangian. Then the lagrangian is,

$$\mathcal{L} = -\frac{1}{2}Z_\phi\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}Z_m m^2\phi^2 - \frac{\lambda}{4!}Z_\lambda\phi^4 \quad (2.43)$$

We have three unknowns¹¹, Z_ϕ, Z_m and Z_λ . We need three conditions to find these. We impose the following,

1. m is the physical mass measured from experiments
2. λ is the cross section measured in some particular scattering experiment
3. $\langle k | \phi(x) | \Omega \rangle = e^{-ikx}$

These are called the *Renormalization Conditions*.

If we want we can re-write this by doing,

$$Z_\phi = 1 + \delta_\phi \quad m^2 Z_m = m^2 + \delta_m \quad \lambda Z_\lambda = \lambda + \delta_\lambda \quad (2.44)$$

Then,

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 + \mathcal{L}_{ct} \quad (2.45)$$

Where $\mathcal{L}_{ct}$ is called the *counter term lagrangian*, and is given by,

$$\mathcal{L}_{ct} = -\frac{1}{2}\delta_\phi\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}\delta_m\phi^2 - \frac{1}{4!}\delta_\lambda\phi^4 \quad (2.46)$$

¹¹Alternatively, m_0, λ_0 and Z

3 Path Integrals in Quantum Mechanics

Path integrals in quantum mechanics was developed by Richard Feynman after getting inspired from a line in Dirac's textbook. The technique is not really useful to study quantum mechanics. But it turns out that it is extremely useful to study quantum field theory. We will “derive” the Feynman Path Integral for quantum mechanics in this chapter. The derivation is rather quick and sloppy, but you can check out any standard book on path integrals for an elaborate version of the same.

3.1 The Propagator

The central question in physics is to find out how a system evolves with time given an initial state. Quantum mechanics is no different. The key question that we wish to find the answer for is this: given that the state of the system at time t_0 is $|\psi(t_0)\rangle$, what would be the state of the system at time $t > t_0$? In order to answer this question, we must specify the rule of evolution of the system. This is given by the Hamiltonian of the system. The Hamiltonian for a single particle system usually has the form,

$$H(\hat{Q}, \hat{P}) = \frac{\hat{P}^2}{2m} + V(\hat{Q}) \quad (3.1)$$

The evolution of the state is then written down in terms of the Hamiltonian as,

$$|\psi(t)\rangle = \exp\left\{-iH(\hat{Q}, \hat{P})(t - t_0)\right\} |\psi(t_0)\rangle \quad (3.2)$$

If we are interested in the position space representation of the state (usually called the Wave function), we project the above equation to position space,

$$\langle q|\psi(t)\rangle = \langle q|\exp\left\{-iH(\hat{Q}, \hat{P})(t - t_0)\right\} |\psi(t_0)\rangle \quad (3.3)$$

Inserting an identity after the exponential,

$$1 = \int dq' |q'\rangle \langle q'| \quad (3.4)$$

we get,

$$\begin{aligned} \psi(q, t) &= \int dq' \langle q|\exp\left\{-iH(\hat{Q}, \hat{P})(t - t_0)\right\} |q'\rangle \langle q'|\psi(t_0)\rangle \\ &= \int dq' G(q, t; q', t_0) \psi(q', t_0) \end{aligned} \quad (3.5)$$

where,

$$G(q, t; q', t_0) := \langle q | \exp \left\{ -iH(\hat{Q}, \hat{P})(t - t_0) \right\} | q' \rangle \quad (3.6)$$

is called the *Propagator*. Now, using the fact that¹²

$$|q, t\rangle = e^{iHt} |q\rangle \quad (3.7)$$

in the Heisenberg picture, we can write,

$$G(q, t; q', t_0) = \langle q, t | q', t_0 \rangle \quad (3.8)$$

This can be interpreted as the probability amplitude for the particle to start from q' and reach at q in a time interval of $t - t_0$.

3.2 Evaluation of the Propagator

Now we find the propagator using the Path integral method. Let $T = t - t_0$. We slice the time interval into $N + 1$ equal intervals with length $\varepsilon = \frac{T}{N+1}$. Then we write,

$$\langle q, T | q', 0 \rangle = \left(\prod_{i=1}^N \int dq_i \right) \langle q_{N+1}, t_{N+1} | q_N, t_N \rangle \langle q_N, t_N | q_{N-1}, t_{N-1} \rangle \cdots \langle q_1, t_1 | q_0, t_0 \rangle \quad (3.9)$$

where $t_{N+1} = T$ and $t_0 = 0$ and so on. Now, take a look at,

$$\begin{aligned} \langle q_{j+1}, t_{j+1} | q_j, t_j \rangle &= \langle q_{j+1} | e^{-i\hat{H}(t_{j+1}-t_j)} | q_j \rangle \\ &= \langle q_{j+1} | e^{-i\hat{H}\varepsilon} | q_j \rangle \\ &= \int dp_j \langle q_{j+1} | p_j \rangle \langle p_j | e^{-i\left(\frac{\hat{P}^2}{2m} + V(\hat{Q})\right)\varepsilon} | q_j \rangle \end{aligned} \quad (3.10)$$

Usually,

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{\frac{1}{2}[\hat{A}, \hat{B}]} \quad (3.11)$$

But in this case, since ε is a small quantity, the commutator term will be e^{ε^2} , and we ignore it. Therefore,

$$\begin{aligned} \langle q_{j+1}, t_{j+1} | q_j, t_j \rangle &= \int dp_j \langle q_{j+1} | p_j \rangle \langle p_j | e^{-i\left(\frac{\hat{P}^2}{2m}\right)\varepsilon} e^{-iV(\hat{Q})\varepsilon} | q_j \rangle \\ &= \int dp_j \langle q_{j+1} | p_j \rangle e^{-i\left(\frac{p_j^2}{2m}\right)\varepsilon} e^{-iV(q_j)\varepsilon} \langle p_j | q_j \rangle \\ &= \int \frac{dp_j}{2\pi} \exp \left\{ -i \left(\frac{p_j^2}{2m} + V(q_j) \right) \varepsilon \right\} \exp \{ i p_j (q_{j+1} - q_j) \} \\ &= \int \frac{dp_j}{2\pi} \exp \left\{ i \left(p_j \frac{q_{j+1} - q_j}{\varepsilon} - H(p_j, q_j) \right) \varepsilon \right\} \end{aligned} \quad (3.12)$$

¹²Remember, basis vectors evolve opposite to the physical states!

where,

$$\langle q|p\rangle = \frac{e^{ipq}}{\sqrt{2\pi}} \quad (3.13)$$

was used. Calling $\frac{q_{j+1}-q_j}{\varepsilon} = \dot{q}_j$ in the limit of $\varepsilon \rightarrow 0$, we have,

$$\langle q, T|q', 0\rangle = \left(\prod_{i=1}^N \int dq_i\right) \left(\prod_{j=0}^N \int \frac{dp_j}{2\pi}\right) \exp\left\{i\varepsilon \sum_{k=0}^N (p_k \dot{q}_k - H(p_k, q_k))\right\} \quad (3.14)$$

In the limit of $\varepsilon \rightarrow 0$, $\varepsilon \sum \rightarrow \int_0^T dt$. Then we have (in fancy notation),

$$\langle q, T|q', 0\rangle = \int \mathcal{D}q \mathcal{D}p \exp\left\{i \int_0^T dt (p(t) \dot{q}(t) - H(p(t), q(t)))\right\} \quad (3.15)$$

Remark : The integral is done with $q(0)$ and $q(T)$ fixed. No such restriction on p .

If the Hamiltonian is at most quadratic in p , then we can do the p integral. To see how, let us use the result,

$$\int_{-\infty}^{\infty} dx e^{f(x)} \approx e^{f(x_0)} \sqrt{\frac{2\pi}{f''(x_0)}} \quad (3.16)$$

Where x_0 is the extremum of f . The way to prove it is to expand f around the extremum point x_0 upto quadratic order and do the gaussian integral. But, when $f(x)$ is at most quadratic in x , this formula is exact. We can do the same thing when H is at most quadratic in p . Find the value of p for which $p\dot{q} - H(p, q)$ is an extremum, substitute that value of p back into $p\dot{q} - H$. i.e,

$$0 = \frac{\partial}{\partial p}(p\dot{q} - H) \implies \dot{q} = \frac{\partial H}{\partial p} \quad (3.17)$$

Solve for p from this and substitute it back. But this is precisely what we do to go from Hamiltonian to Lagrangian.

$$p\dot{q} - H(p, q) \Big|_{p \text{ from } \dot{q} = \partial H / \partial p} = L(q, \dot{q}) \quad (3.18)$$

Hence we end up with,

$$\langle q, T|q', 0\rangle = \int \mathcal{D}q \exp\left\{i \int_0^T dt L(q, \dot{q})\right\} \quad (3.19)$$

The constants have been absorbed to the measure.

3.3 Generalizations

What would be, $\langle q, T | \hat{Q}(t_a) | q', 0 \rangle$? We perform the same song and dance as we did before and we insert identities for each time step. We make sure to insert the identity involving $|q_a, t_a\rangle$ near $\hat{Q}(t_a)$. Then $\hat{Q}(t_a)$ becomes the eigen value $q(t_a)$. Then the rest is the same, and we will have,

$$\langle q, T | \hat{Q}(t_a) | q', 0 \rangle = \int \mathcal{D}q q(t_a) e^{iS[q]} \quad (3.20)$$

What about, $\langle q, T | \hat{Q}(t_a) \hat{Q}(t_b) | q', 0 \rangle$? We can try to play the same game. Notice that in the derivation of the propagator, the identity operators that we inserted at each time step is such that the latest times appear to the left. Therefore, if $t_a < t_b$, then we cannot find the above quantity in the same way. But if $t_a > t_b$, then we can do the same thing and end up with,

$$\langle q, T | \hat{Q}(t_a) \hat{Q}(t_b) | q', 0 \rangle = \int \mathcal{D}q q(t_a) q(t_b) e^{iS[q]} \quad t_a > t_b \quad (3.21)$$

Inside the functional integral, the order doesn't matter since the quantities are not operators. But, we will get the same result for $\langle q, T | \hat{Q}(t_b) \hat{Q}(t_a) | q', 0 \rangle$ if $t_b > t_a$. Hence, we conclude,

$$\int \mathcal{D}q q(t_a) q(t_b) e^{iS[q]} = \begin{cases} \langle q, T | \hat{Q}(t_a) \hat{Q}(t_b) | q', 0 \rangle & \text{if } t_a > t_b \\ \langle q, T | \hat{Q}(t_b) \hat{Q}(t_a) | q', 0 \rangle & \text{if } t_b > t_a \end{cases} \quad (3.22)$$

Define the *time ordering operator*,

$$\mathcal{T}(\hat{Q}(t_a) \hat{Q}(t_b)) := \hat{Q}(t_a) \hat{Q}(t_b) \theta(t_a - t_b) + \hat{Q}(t_b) \hat{Q}(t_a) \theta(t_b - t_a) \quad (3.23)$$

Then we have,

$$\langle q, T | \mathcal{T}(\hat{Q}(t_a) \hat{Q}(t_b)) | q', 0 \rangle = \int \mathcal{D}q q(t_a) q(t_b) e^{iS[q]} \quad (3.24)$$

Now, we invent a trick that let's us generate these time ordered expectation values really quickly. Define,

$$\langle q, T | q', 0 \rangle_f = \int \mathcal{D}q \exp \left\{ i \int_0^T (L + f(t) q(t)) \right\} \quad (3.25)$$

Using the definition of a functional derivative,

$$\frac{\delta f(t)}{\delta f(t')} = \delta(t - t') \quad (3.26)$$

We get,

$$\int \mathcal{D}q q(t_a) e^{iS[q]} = \int \mathcal{D}q \left(\frac{1}{i} \frac{\delta}{\delta f(t_a)} \exp \left\{ i \int_0^T (L + f(t) q(t)) \right\} \right) \Big|_{f=0} = \frac{1}{i} \frac{\delta}{\delta f(t_a)} \langle q, T | q', 0 \rangle_f \Big|_{f=0} \quad (3.27)$$

In general,

$$\langle q, T | \mathcal{T}(\hat{Q}(t_a)\hat{Q}(t_b)\cdots) | q', 0 \rangle = \frac{1}{i} \frac{\delta}{\delta f(t_a)} \frac{1}{i} \frac{\delta}{\delta f(t_b)} \cdots \langle q, T | q', 0 \rangle_f \Big|_{f=0} \quad (3.28)$$

The same thing can be done for p .

3.4 From Vacuum To Vacuum

From the LSZ formula, we know that we are interested in the vacuum to vacuum expectation value of time ordered operators. Let us develop that in terms of path integrals. Let $|0, t\rangle$ be the vacuum a.k.a. the ground state of the Hamiltonian. We are interested in $\langle 0, \infty | 0, -\infty \rangle$. Now in general,

$$\langle 0, t_f | 0, t_i \rangle = \int_{-\infty}^{\infty} dq_f dq_i \langle 0, t_f | q_f, t_f \rangle \langle q_f, t_f | q_i, t_i \rangle \langle q_i, t_i | 0, t_i \rangle \quad (3.29)$$

This is rather complicated. So we pull a little trick. First, we re-scale the Hamiltonian so that the vacuum has energy $E_0 = 0$. Now,

$$\langle q_f, t_f | q_i, t_i \rangle = \langle q_f | e^{-iH(t_f-t_i)} | q_i \rangle = (\langle q_f | e^{-iHt_f}) (e^{iHt_i} | q_i \rangle) \quad (3.30)$$

Consider,

$$e^{iHt_i} | q_i \rangle = e^{iHt_i} \sum_n | n, t_i \rangle \langle n, t_i | q_i \rangle \quad (3.31)$$

$| n \rangle$ are the energy eigen state of the Hamiltonian. To pick out the ground state only, we change the Hamiltonian $H \rightarrow H(1 - i\epsilon)$. Then we have,

$$e^{iH(1-i\epsilon)t_i} | q_i \rangle = \sum_n e^{iE_n(1-i\epsilon)t_i} | n, t_i \rangle \psi_n^*(q_i) \quad (3.32)$$

Now, if we take $t_i \rightarrow -\infty$, only the ground state will be selected. All the other terms in the sum will go to zero due to the small exponential factor, but $E_0 = 0$ will save the ground state term. Therefore,

$$\lim_{t_i \rightarrow -\infty} e^{iH(1-i\epsilon)t_i} | q_i \rangle = | 0, -\infty \rangle \psi_0^*(q_i) \quad (3.33)$$

By the same token,

$$\lim_{t_f \rightarrow \infty} \langle q_f | e^{-iH(1-i\epsilon)t_f} = \langle 0, \infty | \psi_0(q_f) \quad (3.34)$$

The above same are true no matter what we put for $q(t_i) = q_i$ or $q(t_f) = q_f$ as boundary conditions. Therefore, we have,

$$\langle 0, \infty | 0, -\infty \rangle \psi_0(q_f) \psi_0^*(q_i) = \lim_{t_i \rightarrow -\infty} \lim_{t_f \rightarrow \infty} \langle q_f | e^{-iH(1-i\epsilon)(t_f-t_i)} | q_i \rangle \quad (3.35)$$

Since the initial and final values of $q_{i,f}$ doesn't matter, we can average over these and still get the same result. Therefore, we integrate over q_i and q_f , and absorb all the numbers after the integration on the left to the measure, we can write,

$$\langle 0, \infty | 0, -\infty \rangle_{f,h} = \int \mathcal{D}q \mathcal{D}p \exp \left\{ i \int_0^T dt (p\dot{q} - (1-i\epsilon)H(p, q) + fq + hp) \right\} \quad (3.36)$$

Remark: In the above expression, both q and p are unrestricted at the boundaries.

4 Path Integral for Free Field Theory

In this section, we apply the method of Path integrals developed in the last section to a real scalar field. In the process, we will develop a pictorial notation for certain mathematical expressions.

4.1 Free Scalar Field Theory

The real scalar field theory is defined by the Lagrangian,

$$\mathcal{L}_0 = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 \quad (4.1)$$

We are working with the mostly plus metric signature.

The corresponding Hamiltonian is,

$$\mathcal{H}_0 = \frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 \quad (4.2)$$

To get the vacuum to vacuum transition amplitude, we use the Feynman prescription. We change $\mathcal{H}_0$ to $\mathcal{H}_0(1 - i\epsilon)$. This change is equivalent to changing the mass term m^2 to $m^2(1 - i\epsilon)$ ¹³. With this change, we set out to compute the vacuum to vacuum transition amplitude in the presence of a source, denoted by,

$$Z_0[J] \equiv \langle 0; \infty | 0; -\infty \rangle := \int \mathcal{D}\phi \exp \left\{ i \int_x -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2(1 - i\epsilon)\phi^2 + J\phi \right\} \quad (4.3)$$

We have introduced a short hand notation.

$$\int_x \equiv \int d^4x \quad (4.4)$$

To compute the path integral, we first rewrite the action in a particular way. First, integrate by parts the first term in the action to get,

$$\int_x -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi = \int_x \frac{1}{2}\phi\partial^2\phi + \text{boundary term} \quad (4.5)$$

where $\partial^2 = \partial_\mu\partial^\mu$. We can safely set the boundary term to zero. Therefore the action can be rewritten as,

$$S = \int_x -\frac{1}{2}\phi(-\partial^2 + m^2(1 - i\epsilon))\phi + J\phi \quad (4.6)$$

¹³Details can be found at chapters 7 & 8 of Srednicki

We can rewrite $m^2(1 - i\epsilon) = m^2 - i\epsilon$.

Let

$$-\partial^2 + m^2 - i\epsilon \equiv \mathfrak{D} \quad (4.7)$$

Then the action can be rewritten as follows¹⁴,

$$S = \int_x \left\{ -\frac{1}{2} \phi \mathfrak{D} \phi + J \phi \right\} = \int_x \left\{ -\frac{1}{2} (\phi - J \mathfrak{D}^{-1}) \mathfrak{D} (\phi - \mathfrak{D}^{-1} J) + \frac{1}{2} J \mathfrak{D}^{-1} J \right\} \quad (4.8)$$

Therefore, the path integral that we have to compute is,

$$\begin{aligned} Z_0[J] &= \int \mathcal{D}\phi \exp \left\{ i \int_x \left\{ -\frac{1}{2} (\phi - J \mathfrak{D}^{-1}) \mathfrak{D} (\phi - \mathfrak{D}^{-1} J) + \frac{1}{2} J \mathfrak{D}^{-1} J \right\} \right\} \\ &= \exp \left\{ \frac{i}{2} \int_x J \mathfrak{D}^{-1} J \right\} \int \mathcal{D}\phi \exp \left\{ i \int_x \left[-\frac{1}{2} \left(\phi - \frac{J}{\mathfrak{D}} \right) \mathfrak{D}_x \left(\phi - \frac{J}{\mathfrak{D}} \right) \right] \right\} \end{aligned} \quad (4.9)$$

Now, we start waving our hands. Define a shift¹⁵,

$$\chi = \phi - \frac{J}{\mathfrak{D}} \quad (4.10)$$

The shift being a constant with respect to ϕ , we get,

$$Z_0[J] = \exp \left\{ \frac{i}{2} \int_x J \mathfrak{D}^{-1} J \right\} \int \mathcal{D}\chi \exp \left\{ i \int_x -\frac{1}{2} \chi \mathfrak{D} \chi \right\} \quad (4.11)$$

Since we are always interested in computing the functional derivatives of $\log Z_0[J]$ with respect to $J(x)$, the overall constant $Z_0[J]$ has doesn't matter. Therefore we can set $Z_0[J = 0] = 1$. This gives,

$$Z_0[J = 0] = \int \mathcal{D}\chi \exp \left\{ i \int_x -\frac{1}{2} \chi \mathfrak{D} \chi \right\} = 1 \quad (4.12)$$

Hence, the path integral result is simply,

$$Z_0[J] = \exp \left\{ \frac{i}{2} \int_x J \mathfrak{D}^{-1} J \right\} \quad (4.13)$$

¹⁴This is just completing the square! Verify this by expanding it out!

¹⁵You might have noticed that we committed a sin here! This will be remedied soon.

Meaning of $\mathfrak{D}^{-1}$

Let's take a step back and understand the meaning of $\mathfrak{D}^{-1}$. Consider the following equation;

$$\mathfrak{D} |f\rangle = |g\rangle \quad (4.14)$$

This can be formally solved as,

$$|f\rangle = \mathfrak{D}^{-1} |g\rangle + |h\rangle \quad (4.15)$$

where $\mathfrak{D} |h\rangle = 0$. Now if we wish to express the result in position space, we have,

$$\begin{aligned} f(x) &\equiv \langle x|f\rangle = \langle x|\mathfrak{D}^{-1}|g\rangle + h(x) \\ &= \int_y \langle x|\mathfrak{D}^{-1}|y\rangle g(y) + h(x) \\ &= \int_y G(x,y) f(y) + h(x) \end{aligned} \quad (4.16)$$

We have inserted $\mathbb{1} = \int_y |y\rangle \langle y|$ in the second step. What is the meaning of $G(x,y)$? To understand that, we act on the equation with $\mathfrak{D}_x$, the position space version of $\mathfrak{D}$. This gives,

$$g(x) = \mathfrak{D}_x f(x) = \int_y (\mathfrak{D}_x G(x,y)) g(y) + 0 \quad (4.17)$$

This tells us,

$$\mathfrak{D}_x G(x,y) = \delta(x-y) \quad (4.18)$$

Therefore we get the following interpretation: $\mathfrak{D}^{-1}$ is just the Green's function of the operator $\mathfrak{D}$! And,

$$\langle x|\mathfrak{D}^{-1}|g\rangle = \int_y G(x,y) g(y) \quad (4.19)$$

Back to $Z_0[J]$

With this insight, we can rewrite the path integral result as,

$$\boxed{Z_0[J] = \exp \left\{ \frac{i}{2} \int_{x,y} J(x) \Delta(x,y) J(y) \right\}} \quad (4.20)$$

where $\Delta(x, y)$ is the Green's function for $\mathfrak{D} = -\partial^2 + m^2 - i\epsilon$

To find the Green's function, we need to solve,

$$(-\square_x + m^2 - i\epsilon)\Delta(x - y) = \delta^4(x - y) \quad (4.21)$$

The argument of the Green's function is $x - y$ because the right hand side and $\mathfrak{D}$ has that shift invariance. Fourier transform both sides to get,

$$(-\square_x + m^2 - i\epsilon) \int \frac{d^4k}{(2\pi)^4} e^{ik(x-y)} \tilde{\Delta}(k) = \int \frac{d^4k}{(2\pi)^4} e^{ik(x-y)} \quad (4.22)$$

Here we have used a short hand notation, $a^\mu b_\mu = ab$. Acting with the derivative operator and rearranging,

$$\int \frac{d^4k}{(2\pi)^4} e^{ik(x-y)} \left[(k^2 + m^2 - i\epsilon) \tilde{\Delta}(k) - 1 \right] = 0 \quad (4.23)$$

Since e^{ikx} are a complete basis, we get,

$$\tilde{\Delta}(k) = \frac{1}{k^2 + m^2 - i\epsilon} \quad (4.24)$$

Or,

$$\boxed{\Delta(x - y) = \int \frac{d^4k}{(2\pi)^4} \frac{e^{ik(x-y)}}{k^2 + m^2 - i\epsilon}} \quad (4.25)$$

Now, let me tell you of a way that we can cleanse the sin we've committed earlier (the ϕ to χ shift!). We can do the shift in Fourier space without any trouble because operators become numbers in Fourier space. This is what you will find if you read through chapters 7 & 8 of Srednicki!

4.2 Blobs, Lines and all that

We now introduce pictures to represent certain quantities. Let,

$$\frac{1}{i} \Delta(x - y) = \text{---}\overset{\bullet}{x}\text{---}\overset{\bullet}{y} \quad (4.26)$$

$$i \int_x J(x) = \bullet \quad (4.27)$$

With these two rules for pictures, we can write, for example,

$$\overset{\bullet}{x} \text{---} \bullet = i \int_y J(y) \frac{1}{i} \Delta(x-y) \quad (4.28)$$

and,

$$\bullet \text{---} \bullet = i \int_x J(x) i \int_y J(y) \frac{1}{i} \Delta(x-y) = i \int_{x,y} J(x) \Delta(x-y) J(y) \quad (4.29)$$

Thus we can rewrite $Z_0[J]$ as,

$$\boxed{Z_0[J] = \exp \left\{ \frac{1}{2} \bullet \text{---} \bullet \right\}} \quad (4.30)$$

4.3 Wick's Theorem

In QFT, we are interested in correlation functions of the form $\langle 0 | T \left(\hat{\phi}(x_1) \hat{\phi}(x_2) \hat{\phi}(x_3) \hat{\phi}(x_4) \right) | 0 \rangle$, where T is the time ordering symbol. The way we compute this is as follows¹⁶,

$$\langle 0 | T \left(\hat{\phi}(x_1) \hat{\phi}(x_2) \hat{\phi}(x_3) \hat{\phi}(x_4) \right) | 0 \rangle = \left(\frac{1}{i} \frac{\delta}{\delta J(x_1)} \right) \left(\frac{1}{i} \frac{\delta}{\delta J(x_2)} \right) \left(\frac{1}{i} \frac{\delta}{\delta J(x_3)} \right) \left(\frac{1}{i} \frac{\delta}{\delta J(x_4)} \right) Z_0[J] \Big|_{J=0} \quad (4.31)$$

This is called a *4 point correlation function*. Let us compute this. We will need to take functional derivatives of pictures. Let's find the following first;

$$\frac{1}{i} \frac{\delta}{\delta J(x)} \bullet \text{---} \bullet = \frac{1}{i} \frac{\delta}{\delta J(x)} i \int_{y,z} J(y) \Delta(y-z) J(z) \quad (4.32)$$

The rule for the functional derivative is,

$$\frac{\delta J(y)}{\delta J(x)} = \delta^4(x-y) \quad (4.33)$$

Using this, we have,

$$\begin{aligned} \frac{1}{i} \frac{\delta}{\delta J(x)} \bullet \text{---} \bullet &= \int_y J(y) \Delta(y-z) \delta^4(x-z) + \int_z \delta^4(y-x) \Delta(y-z) J(z) \\ &= 2 \int_y J(y) \Delta(y-x) \\ &= 2 \overset{\bullet}{x} \text{---} \bullet \end{aligned} \quad (4.34)$$

¹⁶Read Chapter 6 of Srednicki for more

Here we have used the fact that $\Delta(x - y) = \Delta(y - x)$ ¹⁷ Also,

$$\begin{aligned}
\frac{1}{i} \frac{\delta}{\delta J(y)} \text{ } \overset{\bullet}{x} \text{---} \bullet &= \frac{1}{i} \frac{\delta}{\delta J(y)} \int_z J(z) \Delta(x - z) \\
&= \frac{1}{i} \Delta(x - y) \\
&= \text{ } \overset{\bullet}{x} \text{---} \overset{\bullet}{y}
\end{aligned} \tag{4.35}$$

So we come up with the following rules to take a functional derivative of a picture:

1. Acting with $\frac{1}{i} \frac{\delta}{\delta J(\textcolor{red}{x})}$, has the effect of removing a blob in the picture and replacing it by the spacetime label $\textcolor{red}{x}$.
2. If the original picture had n blobs, do step 1 for all blobs and add all the results up¹⁸

Now armed with these new rules, let's compute the 4 point correlation function. To save writing, we introduce the following short hand,

$$\frac{1}{i} \frac{\delta}{\delta J(\textcolor{red}{x}_k)} \equiv \delta_{\textcolor{red}{k}} \tag{4.36}$$

Now we have to take four derivatives of $Z_0[J]$. $Z_0[J]$ can be formally expanded as,

$$Z_0[J] = 1 + \frac{1}{2} \bullet \text{---} \bullet + \frac{1}{2!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 + \frac{1}{3!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^3 + \dots \tag{4.37}$$

The third term in the expansion has 4 blobs(it is blob-line-blob squared). Therefore, if we take four derivatives of the expansion, the first two terms will go to zero since they don't have four blobs. Terms above the third term will have left over blobs after taking four derivatives. So in the end when we set $J = 0$, all these terms will also vanish! So the only relevant term is the third term. Hence we have,

$$\langle 0 | T \left(\hat{\phi}(x_1) \hat{\phi}(x_2) \hat{\phi}(x_3) \hat{\phi}(x_4) \right) | 0 \rangle = \delta_1 \delta_2 \delta_3 \delta_4 \frac{1}{2!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \tag{4.38}$$

Let us slowly take all the derivatives. First,

¹⁷This is quite clear from the fourier expression 4.25. Swapping x and y seems to make a change in 4.25, but all we have to do is to change k to $-k$ to get the expression back.

¹⁸This is just the Leibniz rule!

$$\begin{aligned}
\delta_4 \frac{1}{2!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 &= \left(\frac{1}{2} \bullet \text{---} \bullet \right) \delta_4 \left(\frac{1}{2} \bullet \text{---} \bullet \right) \\
&= \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right)
\end{aligned} \tag{4.39}$$

Now,

$$\begin{aligned}
\delta_3 \delta_4 \frac{1}{2!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 &= \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right) \delta_3 \left(\frac{1}{2} \bullet \text{---} \bullet \right) + \left(\frac{1}{2} \bullet \text{---} \bullet \right) \delta_3 \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right) \\
&= \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right) \left(\begin{array}{c} \bullet \\ \text{---} x_3 \end{array} \bullet \right) + \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_3 \text{---} x_4 \end{array} \right)
\end{aligned} \tag{4.40}$$

Taking the next derivative,

$$\begin{aligned}
\delta_2 \delta_3 \delta_4 \frac{1}{2!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 &= \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right) \delta_2 \left(\begin{array}{c} \bullet \\ \text{---} x_3 \end{array} \bullet \right) + \left(\begin{array}{c} \bullet \\ \text{---} x_3 \end{array} \bullet \right) \delta_2 \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right) \\
&\quad + \left(\begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_3 \text{---} x_4 \end{array} \right) \delta_2 \left(\frac{1}{2} \bullet \text{---} \bullet \right) \\
&= \left(\begin{array}{c} \bullet \\ \text{---} x_4 \end{array} \bullet \right) \left(\begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_2 \text{---} x_3 \end{array} \right) + \left(\begin{array}{c} \bullet \\ \text{---} x_3 \end{array} \bullet \right) \left(\begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_2 \text{---} x_4 \end{array} \right) \\
&\quad + \left(\begin{array}{c} \bullet \\ \text{---} x_2 \end{array} \bullet \right) \left(\begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_3 \text{---} x_4 \end{array} \right)
\end{aligned} \tag{4.41}$$

Taking the last derivative,

$$\delta_1 \delta_2 \delta_3 \delta_4 \frac{1}{2!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 = \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_1 \text{---} x_2 \end{array} \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_3 \text{---} x_4 \end{array} + \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_1 \text{---} x_3 \end{array} \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_2 \text{---} x_4 \end{array} + \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_1 \text{---} x_4 \end{array} \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_2 \text{---} x_3 \end{array} \tag{4.42}$$

Hence we have,

$$\langle 0 | T \left(\hat{\phi}(x_1) \hat{\phi}(x_2) \hat{\phi}(x_3) \hat{\phi}(x_4) \right) | 0 \rangle = \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_1 \text{---} x_2 \end{array} \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_3 \text{---} x_4 \end{array} + \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_1 \text{---} x_3 \end{array} \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_2 \text{---} x_4 \end{array} + \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_1 \text{---} x_4 \end{array} \begin{array}{c} \bullet \text{---} \bullet \\ \text{---} x_2 \text{---} x_3 \end{array} \tag{4.43}$$

This is called the *Wick's Theorem*¹⁹

¹⁹This is Wick's theorem for four fields. Wick's theorem to 2n fields is readily generalized

5 Calculus of Pictures : Feynman Rules

In this section, we will develop Path integral formalism for an interacting theory. The pictorial notation that we introduces in the last section will be used for most of the calculations. They turn out to be the famous Feynman Diagrams.

5.1 Interacting Scalar Field Theory

An interacting field theory is given by a lagrangian of the form,

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 \quad (5.1)$$

Where $\mathcal{L}_0$ is the free theory lagrangian that we wrote down before, and $\mathcal{L}_1$ is the interacting part. We would like to compute,

$$Z[J] = \int \mathcal{D}\phi \exp \left\{ i \int_x \mathcal{L}_0 + \mathcal{L}_1 + J\phi \right\} \quad (5.2)$$

It would have been nice if we can reuse the results from the free theory partition function! For that, we notice the following²⁰,

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi \exp \left\{ i \int_y \mathcal{L}_1(\phi(y)) \right\} \exp \left\{ i \int_x (\mathcal{L}_0 + J\phi) \right\} \\ &= \exp \left\{ i \int_y \mathcal{L}_1 \left(\frac{1}{i} \frac{\delta}{\delta J(y)} \right) \right\} \int \mathcal{D}\phi \exp \left\{ \int_x (\mathcal{L}_0 + J\phi) \right\} \\ &= \exp \left\{ i \int_y \mathcal{L}_1 \left(\frac{1}{i} \frac{\delta}{\delta J(y)} \right) \right\} Z_0[J] \end{aligned} \quad (5.3)$$

Now, we consider one very important interaction, called the ϕ^4 (phi four) theory.

5.2 ϕ^4 Theory

We take,

$$\mathcal{L}_1 = -\frac{\lambda}{4!} \phi^4(x) \quad (5.4)$$

²⁰If the second step is not clear, By expanding e^ϕ and $e^{\delta x}$, convince yourself that

$$\int \mathcal{D}\phi \exp\{\phi(x)\} \exp \left\{ i \int_y (\mathcal{L}_0 + J\phi) \right\} = \exp \left\{ \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right) \right\} \int \mathcal{D}\phi \exp \left\{ \int_y (\mathcal{L}_0 + J\phi) \right\}$$

Then we need to compute,

$$Z[J] = \exp \left\{ \frac{-i\lambda}{4!} \int_x \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right)^4 \right\} Z_0[J] \quad (5.5)$$

First, we formally expand both exponentials²¹.

$$Z[J] = \sum_{V=0}^{\infty} \frac{1}{V!} \left(\frac{-i\lambda}{4!} \int_x \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right)^4 \right)^V \sum_{P=0}^{\infty} \frac{1}{P!} \left(\frac{i}{2} \int_{y,z} J(y) \Delta(y-z) J(z) \right)^P \quad (5.6)$$

This is a nasty expression! There are $4V$ functional derivatives and $2P$ sources. So when the dust settles, there will be $E = 4P - 2V$ sources left. Let us analyse some simple cases by fixing values for V and P .

Case $V = 0, P = 1$

This case is pretty trivial. We get,

$$D_{01} = \frac{1}{2} \bullet \text{---} \bullet \quad (5.7)$$

Case $V = 1, P = 3$

$$\begin{aligned} D_{13} &= \frac{-i\lambda}{4!} \int_x (\delta_x)^4 \left\{ \frac{1}{3!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^3 \right\} \\ &= \frac{-i\lambda}{4!} \frac{1}{3!} \int_x (\delta_x)^3 \left\{ 3 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \left(\overset{\cdot}{x} \text{---} \bullet \right) \right\} \\ &= \frac{-i\lambda}{4!} \frac{1}{3!} \int_x (\delta_x)^2 \left\{ 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ x \end{array} \right) + 3 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \left(\overset{\cdot}{x} \text{---} \overset{\cdot}{x} \right) \right\} \end{aligned} \quad (5.8)$$

Since the last propagator starts and ends at the same spacetime point, we denote it as,

$$\overset{\cdot}{x} \text{---} \overset{\cdot}{x} = \begin{array}{c} \bigcirc \\ \cdot \\ x \end{array} \quad (5.9)$$

²¹Lest we getting confused, $(\int_x f(x))^V = \int_{x_1} f(x_1) \cdot \int_{x_2} f(x_2) \cdots \int_{x_V} f(x_V)$

Thus,

$$\begin{aligned}
D_{13} &= \frac{-i\lambda}{4!} \frac{1}{3!} \int_x (\delta_x)^2 \left\{ 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ x \end{array} \right) + 3 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \left(\begin{array}{c} \bigcirc \\ x \end{array} \right) \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{3!} \int_x \delta_x \left\{ 3 \cdot 2 \cdot \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \\ | \\ x \end{array} \right) + 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) 2 \cdot \left(\begin{array}{c} \bigcirc \\ | \\ \bullet \\ | \\ x \end{array} \right) + 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bigcirc \\ | \\ \bullet \\ | \\ x \end{array} \right) \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{3!} \int_x \delta_x \left\{ 3 \cdot 2 \cdot \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \\ | \\ x \end{array} \right) + 3 \cdot 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bigcirc \\ | \\ \bullet \\ | \\ x \end{array} \right) \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{3!} \int_x \left\{ 3 \cdot 2 \cdot 3 \left(\begin{array}{c} \bullet \quad \bigcirc \quad \bullet \\ | \quad | \\ x \end{array} \right) + 3 \cdot 3 \cdot 2 \left(\begin{array}{c} \bullet \quad \bigcirc \quad \bullet \\ | \quad | \\ x \end{array} \right) + 3 \cdot 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bigcirc \quad \bigcirc \\ | \quad | \\ x \end{array} \right) \right\}
\end{aligned} \tag{5.10}$$

Collecting same terms, and cancelling factors, we get,

$$D_{13} = \frac{-i\lambda}{2} \int_x \left(\frac{1}{2} \bullet \text{---} \begin{array}{c} \bigcirc \\ | \\ x \end{array} \bullet \right) + \frac{-i\lambda}{8} \int_x \left(\frac{1}{2} \bullet \text{---} \bullet \right) \begin{array}{c} \bigcirc \quad \bigcirc \\ | \quad | \\ x \end{array} \tag{5.11}$$

Case V = 1, P = 4

$$\begin{aligned}
D_{14} &= \frac{-i\lambda}{4!} \int_x (\delta_x)^4 \left\{ \frac{1}{4!} \left(\frac{1}{2} \bullet \text{---} \bullet \right)^4 \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{4!} \int_x (\delta_x)^3 \left\{ 4 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^3 \left(\begin{array}{c} \bullet \\ | \\ x \end{array} \text{---} \bullet \right) \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{4!} \int_x (\delta_x)^2 \left\{ 4 \cdot 3 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ x \end{array} \right) + 4 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^3 \left(\begin{array}{c} \bigcirc \\ x \end{array} \right) \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{4!} \int_x \delta_x \left\{ 4 \cdot 3 \cdot 2 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \\ | \\ x \end{array} \right) + 4 \cdot 3 \cdot (2+1) \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \left(\begin{array}{c} \bigcirc \\ | \\ \bullet \\ | \\ x \end{array} \right) \right\} \\
&= \frac{-i\lambda}{4!} \frac{1}{4!} \int_x \left\{ 4! \left(\begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \quad \bullet \\ \diagup \quad \diagdown \\ x \end{array} \right) + 4!(3+3) \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right) \left(\begin{array}{c} \bullet \quad \bigcirc \quad \bullet \\ | \quad | \\ x \end{array} \right) + 4 \cdot 3 \cdot 3 \cdot \left(\frac{1}{2} \bullet \text{---} \bullet \right)^2 \begin{array}{c} \bigcirc \quad \bigcirc \\ | \quad | \\ x \end{array} \right\}
\end{aligned} \tag{5.12}$$

In the last two steps, similar terms were combined together. Cancelling factors,

$$D_{14} = -i\lambda \int_x \frac{1}{4!} \text{ (four-point vertex)} + \frac{-i\lambda}{2} \int_x \left(\frac{1}{2} \text{ (two-point vertex)} \right) \left(\frac{1}{2} \text{ (two-point vertex with loop)} \right) + \frac{-i\lambda}{2 \cdot 8} \int_x \left(\frac{1}{2} \text{ (two-point vertex)} \right)^2 \text{ (two-point vertex with loop)} \quad (5.13)$$

We now introduce a new rule for the pictures. We will omit writing the spacetime label at the vertex if it is integrated over. And, we will omit writing the integral symbol and $i\lambda$ as well. So whenever we see a vertex with unspecified spacetime label, we understand it as $i\lambda \int_x$. With this, we have a total of three rules. Let's list them here;

$$1. \quad \text{---} \quad = \quad \frac{1}{i} \Delta(x - y)$$

$$2. \quad \bullet \quad = \quad i \int_x J(x)$$

$$3. \quad \times \quad = \quad -i\lambda \int_x$$

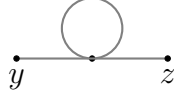
5.3 Symmetry Factors

The pictures in D_{13} will be important when we compute the 2 point correlation function for the full interacting theory precisely because these pictures have two source. If we are computing $\langle 0 | T(\hat{\phi}(x_1) \hat{\phi}(x_2)) | 0 \rangle = \delta_1 \delta_2 Z[J] \Big|_{J=0}$, then taking one functional derivative will neatly cancel the factor of $\frac{1}{2}$ that is tacked on with both diagrams in these pictures. If we look close, we can see that the $\frac{1}{3!}$ term cancelled against a $3!$ that came from the power $P = 3$.

On the other hand, D_{14} will be important when we compute the 4 point correlation function. In the first of these pictures, the $\frac{1}{4!}$ factor will go away when we take four functional derivatives. Like in the previous case, the other $\frac{1}{4!}$ factor was cancelled from $4!$ coming from $P = 4$. In the next term when we take four derivatives, the $\frac{1}{2}$ factors with the pictures neatly cancel, and we'll be left with an overall half factor. In the third picture, the factors of half that is together with the blob diagram squared will go when we take functional derivatives. And then a factor of $\frac{1}{16}$ will be left over.

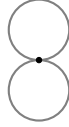
The leftover factors are called the Symmetry factors. There is a way to figure it out without doing all these calculations. Since I have not understood this properly, I'll refrain

from trying to explain this method. Instead, we will look at another way²² of finding the factor. Let's consider the following diagram,



Let us suppose that the blobs are stripped away in a calculation of 2 point correlation functions and they are labelled by y and z . The propagators in the diagram are, $y \rightarrow x$, then $x \rightarrow x$, then $x \rightarrow z$. If we write these one after the other as, $yxxxz$, we can find the number of ways we can create these propagators. Since all the x 's are the same, y can connect/propagate to x in 4 ways. Once we make one $y \rightarrow x$ connection, z can connect to x in 3 ways. There are two x 's left, and they have to connect, which can happen in 1 way only. So we get $4 \cdot 3$ ways of making these connections. Now, each vertex comes with a factor of $\frac{1}{4!}$. Therefore the symmetry factor for the diagram is $\frac{4 \times 3}{4!} = \frac{1}{2}$.

Let's try one more example. Consider,



The propagator sequence can be written as, $x \rightarrow x$ and $x \rightarrow x$ or $xxxx$. The first x connect to another x in three ways. Once that connection is done, the other two x 's has to connect. Therefore the symmetry factor for the diagram is $\frac{3}{4!} = \frac{1}{8}$. But hold on! The third term in D_{14} had a $\frac{1}{2 \times 8}$ factor! What accounts for the $\frac{1}{2}$? This is coming from having two propagators. This will be explained in the next section.

5.4 Connected Diagrams and Vacuum Bubbles

A diagram is said to be connected if we can trace a path from any point in the diagram to any other point without lifting the pen. In general, any term in Z is a product of connected diagrams. Let us enumerate all the countably infinite possible diagrams. Let C_I be connected diagram number I , where $I \in \mathbb{N}$. C_I includes the associated symmetry factor of that particular diagram. To make this clear, let us take a look at such an enumeration,

$$\left\{ C_1 = \frac{1}{4} \text{ (blob with a loop) }, C_2 = \frac{1}{8} \text{ (two loops sharing a vertex) }, C_3 = \frac{1}{4!} \text{ (four blobs connected in a star) }, \dots \right\} \quad (5.14)$$

²²It comes from Wick contraction method. Read more about it in Peskin Chapter 4

Then for this particular enumeration, any term in Z is given by a set of numbers,

$$\{n_I\} = \{n_1, n_2, n_3, \dots\} \quad (5.15)$$

Let me make this clear. Consider a term,

$$C_1^{108} \cdot C_3^{17} \cdot C_{2022}^{93} \quad (5.16)$$

For the chosen enumeration of diagrams, this term is uniquely given by the set,

$$\{n_I\} = \{n_1 = 108, n_2 = 0, n_3 = 17, n_4 = 0, \dots, n_{2021} = 0, n_{2022} = 93, n_{2023} = 0, \dots\} \quad (5.17)$$

Therefore, any general term in Z , denoted by $D_{\{n_I\}}$ can be written as,

$$D_{\{n_I\}} = \frac{1}{S_D} \prod_I (C_I)^{n_I} \quad (5.18)$$

n_I is the number of connected diagrams of type C_I . S_D is an additional symmetry factor on top of whatever factor C_I has. This is coming from repetition of the same diagram in a term. To understand this, let's take a concrete example. Consider,

$$\left(\frac{1}{4} \bullet \text{---} \bigcirc \text{---} \bullet \right)^2 \propto \int_{x,y} \frac{1}{4} \bullet \text{---} \bigcirc \text{---} \bullet \times \frac{1}{4} \bullet \text{---} \bigcirc \text{---} \bullet$$

Notice that we have included the symmetry factor associated with the connected diagram already. The exchange of propagators within the same diagram is accounted for by this factor. But there are still rearrangements possible, which are swapping propagators from one connected diagram to the other. This in general results in inequivalent diagrams unless we swap all the propagators from one diagram with propagators in the other diagram. This swapping results in exactly the same diagram as swapping say the vertices²³, hence we have to account for it. We don't have to count the rearrangements twice. Swapping vertices can be done in $2!$ ways, swapping all propagators from one diagram to the other can also be done in $2!$ ways because this is just permuting the pictures. One of these will be left over in the denominator. Therefore we should have an additional factor of $\frac{1}{2!}$. In general, $(C_I)^{n_I}$ will have an additional symmetry factor $n_I!$. Therefore, the symmetry factor $S_D = \prod_I n_I!$. Hence a general term in Z is,

$$D_{\{n_I\}} = \prod_I \frac{(C_I)^{n_I}}{n_I!} \quad (5.19)$$

²³Which are integrated over hence the label is dummy

Then, we can write Z to be the sum of $D_{\{n_I\}}$'s over all possible sets $\{n_I\}$.

$$\begin{aligned}
Z[J] &\propto \sum_{\{n_I\}} D_{\{n_I\}} \\
&\propto \sum_{\{n_I\}} \prod_I \frac{(C_I)^{n_I}}{n_I!} \\
&\propto \sum_{\{n_I\}} \frac{(C_1)^{n_1}}{n_1!} \frac{(C_2)^{n_2}}{n_2!} \dots
\end{aligned} \tag{5.20}$$

Now, sum over all possible sets²⁴ $\{n_1, n_2, \dots\}$ with each $n_I \in \mathbb{N}$ is equivalent to summing over all n_I separately. Hence,

$$\begin{aligned}
Z &\propto \left(\sum_{n_1} \frac{(C_1)^{n_1}}{n_1!} \right) \left(\sum_{n_2} \frac{(C_2)^{n_2}}{n_2!} \right) \dots \\
&\propto \prod_I \sum_{n_I} \frac{(C_I)^{n_I}}{n_I!} \\
&\propto \prod_I \exp\{C_I\} \\
&\propto \exp\left\{ \sum_I C_I \right\}
\end{aligned} \tag{5.21}$$

Thus, Z is just the exponential of the sum of connected diagrams only!²⁵ We don't have to worry about products of disconnected diagrams at all! Now, we impose the normalization condition, $Z[J=0] = 1$. This gives,

$$\exp\left\{ \sum (\text{diagrams with no sources}) \right\} = 1 \tag{5.22}$$

Define,

$$iW[J] := \sum_{I \neq \{0\}} C_I \tag{5.23}$$

where $I \neq \{0\}$ means we omit all diagrams without sources. Then,

$$\boxed{Z[J] = \exp\{iW[J]\}} \tag{5.24}$$

²⁴Is this sum countably many? In other words, is there an isomorphism $f : \mathbb{N} \times \mathbb{N} \times \dots \rightarrow \mathbb{N}$? There is! $f(n_1, n_2, n_3, \dots) := 2^{n_1} 3^{n_2} 5^{n_3} \dots$ does the job since there are infinitely many primes!

²⁵We'll never have to worry about the additional symmetry factors S_D ever!

with

$$iW[J = 0] = 0 \tag{5.25}$$

6 Feynman Rules in Momentum Space

It is most useful to have Feynman rules in momentum space to perform all the calculations. We now have all the ingredients to derive these. We use the pictures we derived and LSZ reduction formula together to derive Feynman rules in momentum space.

6.1 The Derivation

We derive the rules from applying LSZ formula to one particular diagram. For example, consider,

$$\begin{array}{c} x_2 \\ \diagup \\ \text{---} \bigcirc \text{---} \\ \diagdown \\ x_1 \end{array} \begin{array}{c} x_3 \\ \diagup \\ \text{---} \bigcirc \text{---} \\ \diagdown \\ x_4 \end{array} = \frac{(i\lambda)^2}{2} \int_{y,z} \frac{1}{i} \Delta(x_1 - y) \frac{1}{i} \Delta(x_2 - y) \left(\frac{1}{i} \Delta(y - z) \right)^2 \frac{1}{i} \Delta(z - x_3) \frac{1}{i} \Delta(z - x_4)$$

(6.1)

This diagram is for a $2 \rightarrow 2$ scattering. So we plug this into the LSZ formula for two incoming and two outgoing particles. Therefore, we need to compute,

$$\begin{aligned} \langle f|i \rangle &= (i)^{2+2} \int_{x_1, x_2, x_3, x_4} e^{ik_1 \cdot x_1} e^{ik_2 \cdot x_2} e^{-ik_3 \cdot x_3} e^{-ik_4 \cdot x_4} \\ &\quad \times (-\square_1 + m^2)(-\square_2 + m^2)(-\square_3 + m^2)(-\square_4 + m^2) \times \begin{array}{c} x_2 \\ \diagup \\ \text{---} \bigcirc \text{---} \\ \diagdown \\ x_1 \end{array} \begin{array}{c} x_3 \\ \diagup \\ \text{---} \bigcirc \text{---} \\ \diagdown \\ x_4 \end{array}
 \end{aligned}$$

(6.2)

Where the subscripts on the boxes denote which variable is being differentiated. Now, using,

$$(-\square_x + m^2)\Delta(x - y) = \delta^4(x - y) \quad (6.3)$$

on the diagrams (or on [6.1](#)), we get,

$$\langle f|i \rangle = \frac{(i\lambda)^2}{2} \int_{y,z} e^{ik_1 y} e^{ik_2 y} e^{-ik_3 z} e^{-ik_4 z} \left(\frac{1}{i} \Delta(y - z) \right) \left(\frac{1}{i} \Delta(z - y) \right) \quad (6.4)$$

The reason why we interchanged on one term y and z is to have a proper convention for the momentum direction when we draw momentum space diagrams. We get exactly the same results even if we use the other expression since they are equal! By writing the propagators this way, we roughly think of the first propagator as “propagating from y to z ” and the second one the opposite.

Now, we use,

$$\Delta(x-y) = \int \frac{d^4k}{(2\pi)^4} \frac{e^{ik(x-y)}}{k^2 + m^2 - i\epsilon} \quad (6.5)$$

Therefore we have,

$$\begin{aligned} \langle f|i \rangle &= \frac{(i\lambda)^2}{2} \int_{y,z} e^{i(k_1+k_2)y} e^{-i(k_3+k_4)z} \frac{1}{i^2} \int \frac{d^4p}{(2\pi)^4} \frac{d^4q}{(2\pi)^4} \left(\frac{e^{ip(y-z)}}{p^2 + m^2 - i\epsilon} \right) \left(\frac{e^{iq(z-y)}}{q^2 + m^2 - i\epsilon} \right) \\ &= \frac{(i\lambda)^2}{2} \int \frac{d^4p}{(2\pi)^4} \frac{d^4q}{(2\pi)^4} \left(\frac{-i}{p^2 + m^2 - i\epsilon} \right) \left(\frac{-i}{q^2 + m^2 - i\epsilon} \right) \int_{y,z} e^{i(k_1+k_2+p-q)y} e^{-i(k_3+k_4+p-q)z} \\ &= \frac{(i\lambda)^2}{2} \int \frac{d^4p}{(2\pi)^4} \frac{d^4q}{(2\pi)^4} \left(\frac{-i}{p^2 + m^2 - i\epsilon} \right) \left(\frac{-i}{q^2 + m^2 - i\epsilon} \right) \delta^4(k_1 + k_2 + p - q) \delta^4(k_3 + k_4 + p - q) \\ &= \frac{(i\lambda)^2}{2} \int \frac{d^4p}{(2\pi)^4} \left(\frac{-i}{p^2 + m^2 - i\epsilon} \right) \left(\frac{-i}{(k_1 + k_2 + p)^2 + m^2 - i\epsilon} \right) \delta^4(k_3 + k_4 + p - (p + k_1 + k_2)) \\ &= \frac{(i\lambda)^2}{2} \int \frac{d^4p}{(2\pi)^4} \left(\frac{-i}{p^2 + m^2 - i\epsilon} \right) \left(\frac{-i}{(k_1 + k_2 + p)^2 + m^2 - i\epsilon} \right) \delta^4(k_3 + k_4 - k_1 - k_2) \end{aligned} \quad (6.6)$$

From this derivation, we read off Feynman rules in momentum space. Denote,

$$\langle f|i \rangle = i\mathcal{A} \delta^4(k_{\text{in}} - k_{\text{out}}) \quad (6.7)$$

We will be writing Feynman rules for $i\mathcal{A}$ and keep in mind that the overall four momentum is conserved. If we look close in the derivation, the propagator squared that came from the two vertices in the diagram gave rise to the two factors $\left(\frac{-i}{(k_1+k_2+p)^2+m^2-i\epsilon} \right)$ and $\left(\frac{-i}{p^2+m^2-i\epsilon} \right)$. The propagators that are between two vertices are called *internal lines*. So, here is the first rule:

For each internal line with momentum p , include a factor $\left(\frac{-i}{p^2+m^2-i\epsilon} \right)$

The lines between an external point(say x_1) and a vertex is called an *external line*. The external lines were stripped off by the $(-\square + m^2)$ operators and they ended up contributing a factor of 1. This process is called *amputation*. So, here is the second rule,

For each external line with momentum p , include a factor 1

As a result of writing the Fourier transform of the internal propagators, we ended up with momentum integrals. One of these momentum integrals was taken care of because of one delta function. Where did these delta functions came from? If we take a look at the second to third step in the above derivation, we see that the delta functions came as a result of

integrating out the position labels y and z at the vertices. The exponential factors conspired to give us delta functions. At the y vertex for example, two of the exponential factors (i.e, $e^{i(k_1+k_2)y}$) came as factors for incoming particles in LSZ formula, and the other two (i.e, $e^{i(p+q)y}$) came from the internal propagator. Overall, these exponentials resulted in a conservation of momentum at that vertex. So, that's the third rule:

Conserve four momentum at each vertex

If we look at the delta function at y vertex, it is $\delta^4(k_1 + k_2 + p - q)$. So, we make up the following convention for momenta:

At a vertex, incoming momenta are taken to be positive

So, at this vertex, k_1, k_2 and p are incoming, and q is outgoing. We see that²⁶ this convention is consistent at the other vertex as well. This prompts us to draw the diagram as,



Now, after we impose momentum conservation at each vertex, there will be an overall delta function which we decided not to write. But even after all the momentum conservations, some internal momenta may be left unspecified. These integrals will not go away like the q integral did. So, that's another rule:

Integrate over unspecified four momenta

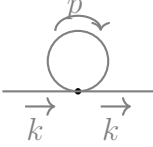
Of course, the vertex factor rule and the symmetry factor rule are inherited from the position space rules. Now, we have all the rules in place. Let's list them;

1. Draw the diagram and mark the direction of momenta. Incoming momenta to a vertex is taken to have a positive sign.
2. External lines with momentum k gets a factor of 1.
3. Internal lines with momentum k gets a factor of $\frac{-i}{k^2+m^2-i\epsilon}$.
4. Each vertex gets a factor $-i\lambda$.

²⁶check this as an exercise

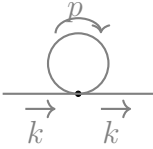
5. Conserve four momentum at each vertex.
6. Integrate over any unspecified momenta.
7. Divide by the symmetry factor.

As an exercise, we look at,



A Feynman diagram representing a tadpole loop. It consists of a horizontal line with two external legs, each labeled with a momentum vector k and an arrow pointing to the right. A vertical line segment connects the midpoint of this horizontal line to a circular loop. The loop has a counter-clockwise arrow and is labeled with a momentum vector p at its top. The entire diagram is labeled (6.9) on the right.

The momentum conservation at the vertex is trivial. We have $k - p + p - k = 0$ trivially. This does not determine the value of p . So we have to integrate over it. The p propagator is an internal line as well. The symmetry factor is just $1/2$. So we get,



$$= \frac{-i\lambda}{2} \int \frac{d^4p}{(2\pi)^4} \frac{-i}{p^2 + m^2 - i\epsilon}$$

The diagram on the left is identical to the one in equation (6.9). The equation is labeled (6.10) on the right.

7 Renormalization of Mass

In this section, we will put everything that we learned so far to use and renormalize the mass of the scalar particle.

7.1 Scalar Self Energy

We define the exact propagator as,

$$\frac{1}{i}\mathcal{G}(x-y) := \langle \Omega | T \phi(x) \phi(y) | \Omega \rangle = \delta_x \delta_y Z[J] \Big|_{J=0} \quad (7.1)$$

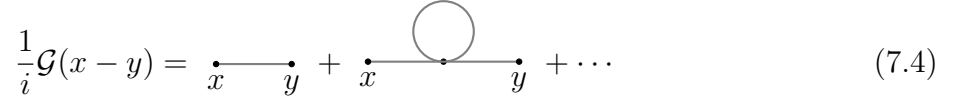
Using,

$$Z[J] = \exp\{iW[J]\} \quad (7.2)$$

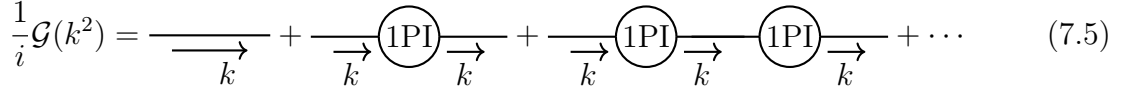
where,

$$iW[J] := \sum_{I \neq \{0\}} C_I \quad (7.3)$$

we can write the exact propagator diagrammatically as,

$$\frac{1}{i}\mathcal{G}(x-y) = \text{---}\overset{\bullet}{x}\text{---}\overset{\bullet}{y}\text{---} + \text{---}\overset{\bullet}{x}\text{---}\text{---}\text{---}\overset{\bullet}{y}\text{---} + \dots \quad (7.4)$$


In the momentum space, all the propagators on the right hand side gets replaced by the momentum space and we have,

$$\frac{1}{i}\mathcal{G}(k^2) = \text{---}\vec{k}\text{---} + \text{---}\vec{k}\text{---}\text{---}\text{---}\vec{k}\text{---} + \text{---}\vec{k}\text{---}\text{---}\text{---}\text{---}\text{---}\vec{k}\text{---} + \dots \quad (7.5)$$


Here, *1PI* stands for one particle irreducible diagrams, which means diagrams if we cut one line, becomes a non-diagram. An example would be the one loop diagram above. Let us denote,

$$-i\Pi(k^2) = \text{---}\vec{k}\text{---}\text{---}\text{---}\vec{k}\text{---} \quad (7.6)$$


Where Π includes only the 1PI part, not the propagators entering and exiting. Then we have,

$$\begin{aligned} \frac{1}{i}\mathcal{G}(k^2) &= \frac{1}{i}\Delta(k^2) + \frac{1}{i}\Delta(k^2) \cdot (-i\Pi(k^2)) \cdot \frac{1}{i}\Delta(k^2) + \frac{1}{i}\Delta(k^2) \cdot (-i\Pi(k^2)) \cdot \frac{1}{i}\Delta(k^2) \cdot (-i\Pi(k^2)) \cdot \frac{1}{i}\Delta(k^2) + \dots \\ &= \frac{1}{i}\Delta(k^2) (1 + (-\Pi\Delta) + (-\Pi\Delta)^2 + \dots) \\ &= \frac{1}{i}\Delta(k^2) \frac{1}{1 + \Pi\Delta} \end{aligned} \quad (7.7)$$

Using,

$$\Delta(k^2) = \frac{1}{k^2 + m^2 - i\epsilon} \quad (7.8)$$

We get the exact propagator to be,

$$\mathcal{G}(k^2) = \frac{1}{k^2 + m^2 + \Pi(k^2) - i\epsilon} \quad (7.9)$$

7.2 Computing $\Pi(k^2)$ to $\mathcal{O}(\lambda)$

We have,

$$\begin{aligned} -i\Pi^{(1)}(k^2) &= \text{Diagram: a circle with a dot at the bottom, two horizontal lines passing through the dot labeled with arrows and 'k', and a curved arrow above the circle labeled 'p'} \\ &= \frac{-i\lambda}{2} \int \frac{d^4p}{(2\pi)^4} \frac{-i}{p^2 + m^2 - i\epsilon} \end{aligned} \quad (7.10)$$

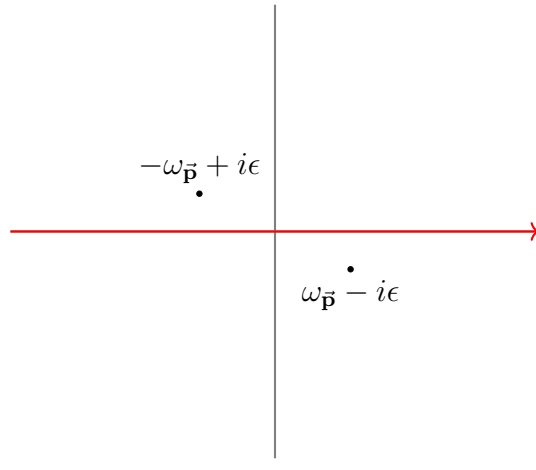
For large p , the integral diverges quadratically. We now perform a series of tricks to extract out the infinity from the integral in a neat way that will be useful in the upcoming sections.

Trick 1 : Wick Rotation

We have,

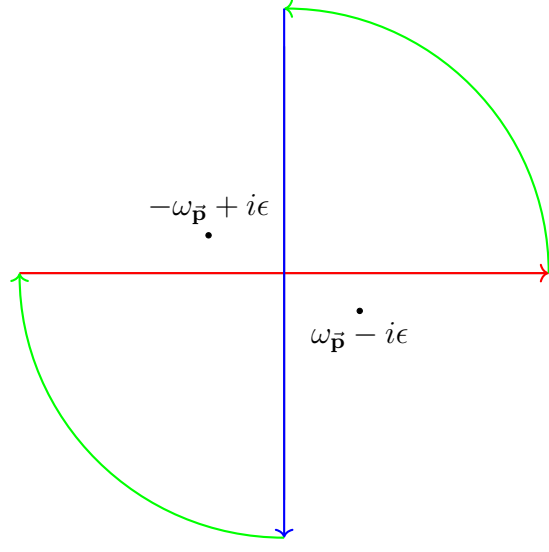
$$I = \int \frac{dp^0 d^3p}{(2\pi)^4} \frac{-i}{-(p^0)^2 + \vec{p}^2 + m^2 - i\epsilon} \quad (7.11)$$

If we take a look at the complex p^0 plane,



$$(7.12)$$

where the red curve shows the integration contour. The role of the $i\epsilon$ are precisely to shift the poles from the integration contour. Now, we can construct the following closed contour, C ,



$$(7.13)$$

Since the region that is bounded by C do not contain any poles, by Cauchy's residue theorem, we have,

$$\oint_C f(p^0) dp^0 = 0 \quad (7.14)$$

for a function f which is analytic in the region. Now, when we push the limits to infinity, the contribution from the green arcs goes to zero. Therefore, we get,

$$0 = \oint_C f(p^0) dp^0 = \int_{-\infty}^{\infty} f(p^0) dp^0 + \int_{i\infty}^{-i\infty} f(p^0) dp^0 \quad (7.15)$$

or,

$$\int_{-\infty}^{\infty} f(p^0) dp^0 = \int_{-i\infty}^{i\infty} f(p^0) dp^0 \quad (7.16)$$

We make the following substitution on the RHS,

$$ip_E^0 = p^0 \quad (7.17)$$

With this substitution, we have

$$I = \int idp^0 \int \frac{d^3p}{(2\pi)^4 + (p_E^0)^2 + \vec{p}^2 + m^2} \frac{-i}{(2\pi)^4 + (p_E^0)^2 + \vec{p}^2 + m^2} \quad (7.18)$$

Notice that we omitted the $i\epsilon$. This is now un-necessary as the integration contour is the vertical line and the poles do not lie on this line! Defining,

$$p_E = (p_E^0, \vec{p}) \quad (7.19)$$

This is just a Euclidian vector. Thus our integral becomes,

$$I = \int \frac{d^4 p_E}{(2\pi)^4} \frac{1}{p_E^2 + m^2} \quad (7.20)$$

with

$$p_E^2 = (p_E^0)^2 + \vec{p}^2 \quad (7.21)$$

Trick 2 : Dimensional Regularization

Since the integral do not converge in 4 dimensions, we do the integral in d dimensions. Then,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \int \frac{d^d p_E}{(2\pi)^d} \frac{1}{p_E^2 + m^2} \quad (7.22)$$

Going to spherical coordinates in d dimensions,

$$\int d^d p_E = \int_0^\infty dp_E p_E^{d-1} \int d\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty dp_E p_E^{d-1} \quad (7.23)$$

Then,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \frac{2\pi^{d/2}}{\Gamma(d/2)} \frac{1}{(2\pi)^d} \int_0^\infty dp_E \frac{p_E^{d-1}}{p_E^2 + m^2} \quad (7.24)$$

Trick 3 : Schwinger Trick

We use the following,

$$\frac{1}{A} = \int_0^\infty dt e^{-tA} \quad (7.25)$$

Therefore we get,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \frac{1}{2^{d-1}\pi^{d/2}} \frac{1}{\Gamma(d/2)} \int_0^\infty dp_E \int_0^\infty dt e^{-t(p_E^2 + m^2)} p_E^{d-1} \quad (7.26)$$

Now, we do the p_E integral.

$$\begin{aligned} \int_0^\infty dp_E e^{-tp_E^2} p_E^{d-1} &= \int_0^\infty d(p_E^2) \frac{1}{2p_E} (p_E^2)^{\frac{d-1}{2}} e^{-tp_E^2} \\ &= \frac{1}{2} \int_0^\infty dy y^{\frac{d}{2}-1} e^{-ty} \quad (y = p_E^2) \\ &= \frac{1}{2} \int_0^\infty \frac{d(ty)}{t} \frac{(ty)^{\frac{d}{2}-1}}{t^{\frac{d}{2}-1}} e^{-ty} \\ &= \frac{1}{2} t^{-d/2} \int_0^\infty du u^{\frac{d}{2}-1} e^{-u} \\ &= \frac{t^{-d/2}}{2} \Gamma(d/2) \end{aligned}$$

Putting this back,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \frac{1}{(4\pi)^{d/2}} \int_0^\infty dt t^{-d/2} e^{-tm^2} \quad (7.27)$$

This integral can also be evaluated using Gamma functions and we get,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \frac{(m^2)^{\frac{d}{2}-1}}{(4\pi)^{d/2}} \Gamma\left(1 - \frac{d}{2}\right) \quad (7.28)$$

Now, we put $d = 4 - \epsilon$, with $0 < \epsilon \ll 1$. This gives us,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \frac{m^2}{16\pi^2} \left(\frac{4\pi}{m^2}\right)^{\epsilon/2} \Gamma\left(-1 + \frac{\epsilon}{2}\right) \quad (7.29)$$

Using the formula,

$$\Gamma(-n + x) = \frac{(-1)^n}{n!} \left(\frac{1}{x} - \gamma + \sum_{k=1}^n \frac{1}{k} + \mathcal{O}(x) \right) \quad (7.30)$$

where γ is just a number. Then,

$$\Gamma\left(-1 + \frac{\epsilon}{2}\right) = -\frac{2}{\epsilon} + (\gamma - 1) + \mathcal{O}(\epsilon) \quad (7.31)$$

Now, we re-wrtie,

$$\left(\frac{4\pi}{m^2}\right)^{\epsilon/2} = \exp\left\{\frac{\epsilon}{2} \ln\left(\frac{4\pi}{m^2}\right)\right\} \quad (7.32)$$

Hmm... Logarithm of a dimensionful quantity? Did we mess up something? Indeed we did! We didn't think carefully about the dimensions of λ . Let us do that now. In d dimensions, the action is,

$$S = \int d^d x \left(-\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda_d}{4!} \phi^4 \right) \quad (7.33)$$

If we use mass as the only dimension full quantity, then S is dimensionless. $d^d x$ has mass dimensions $-d$. Therefore the lagrangian must have mass dimensions d . This gives us the dimensions of the field.

$$2[\phi] + 2 = d \quad (7.34)$$

or,

$$[\phi] = \frac{d-2}{2} \quad (7.35)$$

Therefore,

$$[\lambda_d] + 4[\phi] = d \quad (7.36)$$

gives,

$$[\lambda_d] = 4 - d = \epsilon \quad (7.37)$$

The coupling constant has a mass dimension now²⁷. Let λ be a dimensionless number and M be an arbitrary mass scale. Write,

$$\lambda_d = \lambda M^\epsilon \quad (7.38)$$

We have to replace λ by λM^ϵ in our calculation. This gives us,

$$\Pi^{(1)}(k^2) = \frac{\lambda}{2} \frac{m^2}{16\pi^2} \left(\frac{4\pi M^2}{m^2} \right)^{\epsilon/2} \left(-\frac{2}{\epsilon} + (\gamma - 1) + \mathcal{O}(\epsilon) \right) \quad (7.39)$$

Now,

$$\left(\frac{4\pi M^2}{m^2} \right)^{\epsilon/2} = \exp \left\{ \frac{\epsilon}{2} \ln \left(\frac{4\pi M^2}{m^2} \right) \right\} = 1 + \frac{\epsilon}{2} \ln \left(\frac{4\pi M^2}{m^2} \right) + \mathcal{O}(\epsilon^2) \quad (7.40)$$

This gives,

$$\Pi^1(k^2) = \frac{\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) - \frac{\lambda m^2}{16\pi^2 \epsilon} + \mathcal{O}(\epsilon) \quad (7.41)$$

7.3 Lehman - Kallen Form of the Exact Propagator

We have defined,

$$\frac{1}{i} \mathcal{G}(x - y) := \langle \Omega | T \phi(x) \phi(y) | \Omega \rangle \quad (7.42)$$

Where ϕ is the full interacting field. While deriving the LSZ formula, we imposed on the fields the following conditions,

$$\langle \Omega | \phi(x) | \Omega \rangle = 0 \quad (7.43)$$

and

$$\langle k | \phi(x) | \Omega \rangle = e^{-ikx} \quad (7.44)$$

We achieved this by shifting and re-scaling the field. Before studying the exact propagator, let's take one more look at the free propagator,

$$\Delta(x - y) = \int \frac{d^d k}{(2\pi)^d} e^{ik(x-y)} \Delta(k^2) \quad (7.45)$$

where,

$$\Delta(k^2) = \frac{1}{k^2 + m^2 - i\epsilon} \quad (7.46)$$

²⁷For $d = 4$, λ is dimensionless

We notice that the free propagator has a pole at $k^2 = -m^2$ in the complex k^2 plane and the residue of the pole is 1. We keep this in our mind and study the exact propagator. The first step is to insert an identity operator,

$$I = |\Omega\rangle \langle \Omega| + \int \widetilde{dk} |k\rangle \langle k| + \sum_n \int \widetilde{dk} |k, n\rangle \langle k, n| \quad (7.47)$$

Here,

$$\widetilde{dk} = \frac{d^{d-1}k}{(2\pi)^{d-1}2\omega} \quad (7.48)$$

with,

1. $\omega = \sqrt{\vec{k}^2 + m^2}$ for the one particle integral
2. $\omega = \sqrt{\vec{k}^2 + M^2}$ with $M \geq 2m$ for the multi-particle integral

The sum over n is schematic. Let us assume $x^0 > y^0$ without loss of any generality and insert this identity operator in between ϕ 's.

$$\begin{aligned} \langle \Omega | \phi(x) \phi(y) | \Omega \rangle &= \langle \Omega | \phi(x) | \Omega \rangle \langle \Omega | \phi(y) | \Omega \rangle + \int \widetilde{dk} \langle \Omega | \phi(x) | k \rangle \langle k | \phi(y) | \Omega \rangle \\ &\quad + \sum_n \int \widetilde{dk} \langle \Omega | \phi(x) | k, n \rangle \langle k, n | \phi(y) | \Omega \rangle \end{aligned} \quad (7.49)$$

Using²⁸,

$$\langle k, n | \phi(x) | \Omega \rangle = e^{-ikx} \langle k, n | \phi(0) | \Omega \rangle \quad (7.50)$$

and the normalization conditions, we get,

$$\langle \Omega | \phi(x) \phi(y) | \Omega \rangle = \int \widetilde{dk} e^{ik(x-y)} + \sum_n \int \widetilde{dk} e^{ik(x-y)} |\langle k, n | \phi(0) | \Omega \rangle|^2 \quad (7.51)$$

We define the following quantity, called the *Spectral Density*,

$$\rho(s) := \sum_n |\langle k, n | \phi(0) | \Omega \rangle|^2 \delta(s - M^2) \quad (7.52)$$

ρ is zero for $s < 4m^2$ and greater than zero otherwise. This let's us rewrite the expression as,

$$\langle \Omega | \phi(x) \phi(y) | \Omega \rangle = \int \widetilde{dk} e^{ik(x-y)} + \int_{4m^2}^{\infty} ds \rho(s) \int \widetilde{dk} e^{ik(x-y)} \quad (7.53)$$

²⁸This was derived in the LSZ section

The subtlety to keep in mind is that in the first term, $k^0 = \sqrt{\vec{k}^2 + m^2}$ whereas, in the second term, $k^0 = \sqrt{\vec{k}^2 + s}$. Here, m is the *physical mass* of the particle that one would measure in an experiment. To get the result for the other time ordering ($y^0 > x^0$), we just have to swap x and y .

$$\langle \Omega | \phi(y) \phi(x) | \Omega \rangle = \int \widetilde{dk} e^{-ik(x-y)} + \int_{4m^2}^{\infty} ds \rho(s) \int \widetilde{dk} e^{-ik(x-y)} \quad (7.54)$$

Thus we can write,

$$i \langle \Omega | T \phi(x) \phi(y) | \Omega \rangle = i\theta(x^0 - y^0) \langle \Omega | \phi(x) \phi(y) | \Omega \rangle + i\theta(y^0 - x^0) \langle \Omega | \phi(y) \phi(x) | \Omega \rangle \quad (7.55)$$

Now we use the identity²⁹,

$$\int \frac{d^d k}{(2\pi)^d} \frac{e^{ik(x-y)}}{k^2 + m^2 - i\epsilon} = i\theta(x^0 - y^0) \int \widetilde{dk} e^{ik(x-y)} + i\theta(y^0 - x^0) \int \widetilde{dk} e^{-ik(x-y)} \quad (7.56)$$

to re-wrtie,

$$i \langle \Omega | T \phi(x) \phi(y) | \Omega \rangle = \int \frac{d^d k}{(2\pi)^d} e^{ik(x-y)} \left[\frac{1}{k^2 + m^2 - i\epsilon} + \int_{4m^2}^{\infty} ds \rho(s) \frac{1}{k^2 + s - i\epsilon} \right] \quad (7.57)$$

Therefore in the momentum space,

$$\mathcal{G}(k^2) = \frac{1}{k^2 + m^2 - i\epsilon} + \int_{4m^2}^{\infty} ds \frac{\rho(s)}{k^2 + s - i\epsilon} \quad (7.58)$$

We make the following observations: the exact propagator also has a simple pole at $k^2 = -m^2$, where m is the physical mass of the particle, and the residue is 1. The residue is 1 because we shifted the field to get the normalization the same as the free particle. So, we can find the value of the shift Z by imposing the exact propagator has a pole at the physical mass squared with residue one! This is what we will be using to find the renormalization parameters δ_m and δ_ϕ .

7.4 Counter Term Diagrams

We had,

$$\mathcal{L}_{ct} = -\frac{1}{2} \delta_\phi \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} \delta_m \phi^2 - \frac{1}{4!} \delta_\lambda \phi^4 \quad (7.59)$$

Doing an integration by parts on the first term,

$$\mathcal{L}_{ct} = \frac{1}{2} \delta_\phi \phi \square \phi - \frac{1}{2} \delta_m \phi^2 - \frac{1}{4!} \delta_\lambda \phi^4 \quad (7.60)$$

²⁹this can be derived using a bit of complex analysis. Maybe I'll include this as an appendix

Finding the diagrams for these counter terms consists of the same procedure, as we discussed in the “Lines and Blobs” section. To get the diagrams for the first term, we look at,

$$\exp\left\{\frac{i\delta_\phi}{2}\int_x\delta_x\Box\delta_x\right\}\exp\left\{\frac{1}{2}\bullet\text{---}\bullet\right\} \quad (7.61)$$

where $\delta_x = \frac{1}{i}\frac{\delta}{\delta J(x)}$. Consider,

$$\frac{i\delta_\phi}{2}\int_x\delta_x\Box\delta_x\frac{1}{2}\left(\frac{1}{2}\bullet\text{---}\bullet\right)^2 = \frac{i\delta_\phi}{2}\int_x\delta_x\Box\left(\overset{\cdot}{x}\text{---}\bullet\right)\left(\frac{1}{2}\bullet\text{---}\bullet\right) \quad (7.62)$$

Now,

$$\begin{aligned} \Box\left(\overset{\cdot}{x}\text{---}\bullet\right) &= \Box\int_y J(y)\Delta(y-x) \\ &= \int_y J(y)\Box\Delta(y-x) \end{aligned} \quad (7.63)$$

Now we take the next δ_x on the diagram with two blobs³⁰ Therefore³¹,

$$\frac{i\delta_\phi}{2}\int_x\delta_x\Box\delta_x\left(\frac{1}{2}\bullet\text{---}\bullet\right)^2 \supseteq \frac{i\delta_\phi}{2}\int_{x,y,z} J(y)(\Box\Delta(y-x))\Delta(x-z)J(z) \quad (7.64)$$

We denote this as a new diagram,

$$\frac{1}{2}\bullet\text{---}\overset{x}{\bullet}\text{---}\bullet \quad (7.65)$$

And assign a vertex factor, $i\delta_\phi\int_x\Box$, and we keep in mind that the box acts only on one of the propagators³². Similarly, the vertex factor for $-\frac{1}{2}\delta_m\phi^2$ is easy to find, it will be just $-i\delta_m\int_x$. So, we combine these two factors into one diagram as,

$$\text{---}\overset{\bullet}{\text{---}}\text{---} = -i\int_x(-\delta_\phi\Box + \delta_m) \quad (7.66)$$

And we keep in mind that the “box” only acts on one of the propagators.

The rule for the counter term four point vertex is also easily derived. It is given by,

$$\text{X} = -i\delta_\lambda\int_x \quad (7.67)$$

³⁰If we take it on the one blob diagram, we get disconnected diagrams, which we already have shown to be irrelevant!

³¹The super set symbol just means “contains”

³²Which propagator the derivative acts doesn’t matter because we can change it via integration by parts

Plugging these into the LSZ formula will give us the momentum space rules. They are also quite easy to see. Let's list all the rules for the theory below:

$$\begin{array}{c} \xrightarrow{k} \\ \hline \end{array} = \frac{-i}{k^2 + m^2 - i\epsilon} \quad (\text{for internal lines}) \quad (7.68)$$

$$\begin{array}{c} \diagup \\ \bullet \\ \diagdown \end{array} = -i\lambda \quad (\text{the vertex}) \quad (7.69)$$

$$\begin{array}{c} \bullet \\ \hline \end{array} = -i(\delta_\phi k^2 + \delta_m) \quad (\text{counter term vertex \#1}) \quad (7.70)$$

$$\begin{array}{c} \diagup \\ \bullet \\ \diagdown \end{array} = -i\delta_\lambda \quad (\text{counter term vertex \#2}) \quad (7.71)$$

7.5 Back to The Scalar Self Energy Problem

We calculated,

$$-i\Pi^1(k^2) = \begin{array}{c} \text{p} \\ \curvearrowright \\ \bullet \\ \xrightarrow{k} \quad \xrightarrow{k} \end{array} \quad (7.72)$$

and obtained,

$$-i\Pi^1(k^2) = \frac{-i\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) + \frac{i\lambda m^2}{16\pi^2\epsilon} \quad (7.73)$$

If we include the counter term vertex also, then we get,

$$-i\Pi^1(k^2) = \frac{-i\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) + \frac{i\lambda m^2}{16\pi^2\epsilon} - i(\delta_\phi k^2 + \delta_m) \quad (7.74)$$

What are the conditions to determine δ_ϕ and δ_m ? We have translated the renormalization conditions we saw in LSZ to the following statement: *The exact propagator has a simple pole at the physical mass, and the residue is one.* The exact propagator is,

$$\mathcal{G}(k^2) = \frac{1}{k^2 + m^2 + \Pi(k^2)} \quad (7.75)$$

We impose the renormalization conditions on this.

7.6 On Shell Renormalization Scheme(OS Scheme)

We demand that m is the physical mass. Then for the propagator to have a simple pole at $k^2 = -m^2$, we must have,

$$\Pi(-m^2) = 0 \quad (7.76)$$

To find the residue, we Taylor expand Π as,

$$\Pi(k^2) = \Pi(-m^2) + (k^2 + m^2)\Pi'(-m^2) + \dots \quad (7.77)$$

Plugging this to the exact propagator gives,

$$\mathcal{G}(k^2) = \frac{1}{(k^2 + m^2)(1 + \Pi'(-m^2) + (k^2 + m^2)\Pi''(-m^2) + \dots)} \quad (7.78)$$

For the residue to be one at $k^2 = -m^2$, we need,

$$\Pi'(-m^2) = 0 \quad (7.79)$$

Upto first order in λ , the second condition 7.79 gives us,

$$\left. \frac{d}{dk^2} \Pi^{(1)}(k^2) \right|_{k^2=-m^2} = \delta_\phi = 0 \quad (7.80)$$

Now, the first conditon given by 7.76 gives,

$$0 = \Pi^{(1)}(-m^2) = \frac{\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) - \frac{\lambda m^2}{16\pi^2 \epsilon} + \delta_m \quad (7.81)$$

Thus, we have,

$$\delta_\phi = 0 + \mathcal{O}(\lambda^2) \quad (7.82)$$

and

$$\delta_m = \frac{\lambda m^2}{16\pi^2 \epsilon} - \frac{\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) + \mathcal{O}(\lambda^2) \quad (7.83)$$

7.7 Minimal Subtraction Scheme(MS Scheme)

In this scheme of renormalization, we use the deltas only to cancel the infinite part of the self energy. In other words, we do not demand that the mass parameter m is the physical mass, denoted by m_{phys} . The renormalization conditions still remain the same. We need the exact propagator to have a simple pole at $k^2 = -m_{phys}^2$ with residue 1. This translates to,

$$\left. (k^2 + m^2 + \Pi(k^2)) \right|_{k^2=-m_{phys}^2} = 0 \quad (7.84)$$

or

$$m_{phys}^2 = m^2 + \Pi(-m_{phys}^2) \quad (7.85)$$

The residue conditon would be,

$$\Pi'(-m_{phys}^2) = 0 \quad (7.86)$$

This immediately gives us,

$$\delta_\phi = 0 + \mathcal{O}(\lambda^2) \quad (7.87)$$

If we use δ_m only to cancel the infinte part, i.e, we choose,

$$\delta_m = \frac{\lambda m^2}{16\pi\epsilon} + \mathcal{O}(\lambda^2) \quad (7.88)$$

The pole condtion becomes,

$$m_{phys}^2 = m^2 + \frac{\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) \quad (7.89)$$

We make the following observation: since m_{phys} is measured in an experiment, it cannot depend on M , the arbitrary mass scale that we introduced in the problem. So the coupling constants m^2 and λ must change as a function of M to ensure the constancy of m_{phys} . This will give us the *Running of the coupling constants*. We will explore this more once we compute the corrections to the vertex.

8 Renormalization of Coupling Constant

In this section, we will take a look at the renormalization of the coupling constant λ in our theory. The techniques used would be almost the same as what we did in the previous chapter. The process that we will be looking at is a scattering process $\phi\phi \rightarrow \phi\phi$.

8.1 $\phi\phi \rightarrow \phi\phi$ Scattering

The amplitude for this process diagrammatically is given by,

$$\begin{aligned}
 -i\mathcal{A}(p_1, p_2 \rightarrow p_3, p_4) = & \text{Diagram 1} + \left(\text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} \right) \\
 & + \text{Diagram 5} \quad (8.1)
 \end{aligned}$$

The diagrams represent the following terms in the amplitude:

- Diagram 1:** A four-point vertex (cross) with incoming momenta p_1, p_2 and outgoing momenta p_3, p_4 .
- Diagram 2:** A loop diagram with a circle. Incoming momenta p_1, p_2 enter from the left, and outgoing momenta p_3, p_4 exit to the right. Internal momenta k and q are shown on the loop.
- Diagram 3:** A loop diagram with a circle, similar to Diagram 2 but with different internal momentum assignments.
- Diagram 4:** A loop diagram with a circle, similar to Diagram 2 but with different internal momentum assignments.
- Diagram 5:** A four-point vertex (cross) with a red dot at the center, representing a counterterm.

Let us denote,

$$\text{Diagram 2} = (-i\lambda)^2 iV(p^2) \quad (8.2)$$

Defining Mandelstam variables,

$$s := (p_1 + p_2)^2 \quad t := (p_1 - p_3)^2 \quad u := (p_1 - p_4)^2 \quad (8.3)$$

We have,

$$-i\mathcal{A} = -i\lambda + (-i\lambda)^2 (iV(s) + iV(t) + iV(u)) - i\delta_\lambda \quad (8.4)$$

Let us now evaluate,

$$(-i\lambda)^2 iV(p^2) = \text{Diagram 2} = \frac{(-i\lambda)^2}{2} \int \frac{d^4 k}{(2\pi)^4} \left(\frac{-i}{k^2 + m^2 - i\epsilon} \right) \left(\frac{-i}{(k+p)^2 + m^2 - i\epsilon} \right) \quad (8.5)$$

Or,

$$V(p^2) = \frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \left(\frac{1}{k^2 + m^2 - i\epsilon} \right) \left(\frac{1}{(k+p)^2 + m^2 - i\epsilon} \right) \quad (8.6)$$

We now need a new trick, called the Feynman trick.

Trick 0 : Feynman Parameters

We can write,

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{(xA + (1-x)B)^2} \quad (8.7)$$

This can be verified trivially by doing the integral and is left as an exercise. With this, we get,

$$\frac{1}{(k^2 + m^2)((k+p)^2 + m^2)} = \int_0^1 dx \frac{1}{(k^2 + 2xk \cdot p + xp^2 + m^2)^2} \quad (8.8)$$

Now we do a shift,

$$\ell = k + xp \quad (8.9)$$

Then we have,

$$V(p^2) = \frac{i}{2} \int_0^1 dx \int \frac{d^4 \ell}{(2\pi)^4} \frac{1}{(\ell^2 + p^2 x(1-x) + m^2 - i\epsilon)^2} \quad (8.10)$$

Call,

$$\Delta = p^2 x(1-x) + m^2 - i\epsilon^2 \quad (8.11)$$

Now we can proceed to do the tricks from the previous chapter, starting with Wick rotation. But hold on! Can we do Wick rotation here? The reason why we could do Wick rotation in the last chapter was that the first and third quadrants in the complex k^0 plane had no poles. The poles in the integrand in the last chapter was in the second and fourth quadrants. But now, the integrand apparently has four poles. Are we sure that they are still in the second and fourth quadrants? Let us investigate. The expression we want to Wick rotate is the ℓ integral. Here the four poles have become two double poles, and the poles are at $\ell^0 = \pm \sqrt{\mathbf{l}^2 + p^2 x(1-x) + m^2} \mp i\epsilon$. The poles do look like to be in second and fourth quadrant. Phew! Wick rotation is safe!

But there is yet another problem. Since p^2 can be negative and greater than m^2 , the pole can be (apart from the $i\epsilon$) imaginary. This in general means some decay is going on, and the rate can be extracted by first sending $p^0 \rightarrow ip_E^0$, then doing the same tricks and taking the imaginary part. We omit this discussion here. Now, let us proceed with Wick rotation,

$$V(p^2) = -\frac{1}{2} \int_0^1 dx \int \frac{d^4 \ell_E}{(2\pi)^4} \frac{1}{(\ell_E^2 + \Delta)^2} \quad (8.12)$$

Now we change the integral to a d dimensional integral. Then we have,

$$V(p^2) = -\frac{M^{2\epsilon}}{2} \frac{2\pi^{d/2}}{\Gamma(d/2)(2\pi)^d} \int_0^1 dx \int_0^\infty d\ell_E \ell_E^{d-1} \frac{1}{(\ell_E^2 + \Delta)^2} \quad (8.13)$$

Where the M is exactly the same M as we discussed in the last chapter, to account for dimensional consistency when we do the theory in d dimensions. Using Schwinger trick,

$$V(p^2) = -\frac{M^{2\epsilon}}{2} \frac{2\pi^{d/2}}{\Gamma(d/2)(2\pi)^d} \int_0^1 dx \int_0^\infty d\ell_E \ell_E^{d-1} \int_0^\infty dt t \exp\{\ell_E^2 + \Delta\} \quad (8.14)$$

Now, all the techniques are the same as last chapter. First we do the ℓ_E integral. Then we do the t integral. Then we end up with the result,

$$V(p^2) = -\frac{M^{2\epsilon}}{2 \cdot (4\pi)^{d/2}} \int_0^1 dx \Gamma\left(2 - \frac{d}{2}\right) \Delta^{(d/2)-2} \quad (8.15)$$

Now using $d = 4 - \epsilon$, we get,

$$V(p^2) = -\frac{M^\epsilon}{32\pi^2} \left(\frac{2}{\epsilon} - \gamma\right) \int_0^1 dx \left(\frac{4\pi M^2}{\Delta}\right)^{\epsilon/2} \quad (8.16)$$

But,

$$\left(\frac{4\pi M^2}{\Delta}\right)^{\epsilon/2} = 1 + \frac{\epsilon}{2} \ln \left\{ \frac{4\pi M^2}{\Delta} \right\} + \mathcal{O}(\epsilon^2) \quad (8.17)$$

Then sending $\epsilon \rightarrow 0$ where ever possible, we end up with,

$$V(p^2) = -\frac{1}{32\pi^2} \left[\left(\frac{2}{\epsilon} - \gamma\right) + \int_0^1 dx \ln \left\{ \frac{4\pi M^2}{m^2 + x(1-x)p^2} \right\} \right] \quad (8.18)$$

8.2 On - Shell Scheme

We need to have a renormalization condition for the coupling constant similar to that of the mass renormalization condition in the last chapter. We make up the following condition: λ is the scattering amplitude measured when the total incoming 3-momentum is zero. This translates to $s = 4m^2$ and $t = u = 0$. Plugging this to the scattering amplitude formula, we have,

$$-i\lambda = -i\lambda + (-i\lambda)^2(iV(4m^2) + iV(0) + iV(0)) - i\delta_\lambda \quad (8.19)$$

This gives us,

$$\delta_\lambda = \lambda^2(V(4m^2) + 2V(0)) \quad (8.20)$$

A better choice of renormalization scheme will be discussed in the next chapter and we will arrive at the running of λ with M .

9 Running of Coupling Constants

We have seen in the previous two chapters how we renormalize the mass and the coupling constant to render the results of scattering amplitudes finite (at least up to one loop order). We have kept our focus on the so called On-Shell scheme of renormalization. For the mass case, we assumed that the parameter m in the lagrangian is the actual mass of the particle. This gives us the renormalization condition that the exact propagator must have a pole at $k^2 = -m^2$ with a residue 1, and gives the self energy to one loop order vanish for all external momenta. Now, we focus on the minimal subtraction scheme, in which we use the counter terms only to cancel the infinite part of the diagrams and that gives us the running of the coupling constants, namely m and λ .

9.1 The *Anomalous Mass Dimension* and Running of m

In a previous chapter, we found the exact propagator as,

$$\mathcal{G}(k^2) = \frac{1}{k^2 + m^2 + \Pi(k^2)} \quad (9.1)$$

Then we evaluated the self energy to one loop order to be,

$$\Pi^{(1)}(k^2) = \frac{\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) - \frac{\lambda m^2}{16\pi^2 \epsilon} + (\delta_\phi k^2 + \delta_m) \quad (9.2)$$

where M is an arbitrary mass scale we introduced. We saw that choosing the parameter m in the lagrangian to be the physical mass yields the self energy to be identically zero to first order in λ . Now, there is no reason for the parameter m in the lagrangian to be the physical mass. This is just one choice of renormalization conditions from many. We now choose to renormalize differently, namely using a *Modified Minimal Subtraction scheme*. We say that the physical mass of the particle is m_{phys} . This tells us that the exact propagator must have a simple pole at $k^2 = -m_{phys}^2$. This condition gives us,

$$(k^2 + m^2 + \Pi(k^2)) \Big|_{k^2 = -m_{phys}^2} = 0 \quad (9.3)$$

or

$$m_{phys}^2 = m^2 + \Pi(-m_{phys}^2) \quad (9.4)$$

Since Π is of $\mathcal{O}(\lambda)$, we can write,

$$m_{phys}^2 = m^2 + \mathcal{O}(\lambda) \quad (9.5)$$

and hence,

$$m_{phys}^2 = m^2 + \Pi^{(1)}(-m^2) + \mathcal{O}(\lambda^2) \quad (9.6)$$

To find the residue, expand,

$$\Pi(k^2) = \Pi(-m_{phys}^2) + (k^2 + m_{phys}^2)\Pi'(-m_{phys}^2) + \dots \quad (9.7)$$

The propagator then is,

$$\begin{aligned} \mathcal{G}(k^2) &= \frac{1}{k^2 + m^2 + \Pi(-m_{phys}^2) + (k^2 + m_{phys}^2)\Pi'(-m_{phys}^2) + \dots} \\ &= \frac{1}{k^2 + m_{phys}^2} \frac{1}{1 + \Pi'(-m_{phys}^2) + \dots} \end{aligned} \quad (9.8)$$

Therefore the residue is,

$$R = \frac{1}{1 + \Pi'(-m_{phys}^2)} = \frac{1}{1 + \delta_\phi} \quad (9.9)$$

We make the following choice: we choose the residue to be 1 or $\delta_\phi = 0$. This gives us,

$$m_{phys}^2 = m^2 + \frac{\lambda m^2}{32\pi^2} \left((\gamma - 1) - \ln \frac{4\pi M^2}{m^2} \right) - \frac{\lambda m^2}{16\pi^2 \epsilon} + \delta_m \quad (9.10)$$

Next, we choose,

$$\delta_m = \frac{\lambda m^2}{16\pi^2 \epsilon} - \frac{\lambda m^2}{32\pi^2} ((\gamma - 1) - \ln 4\pi) \quad (9.11)$$

This gives us,

$$m_{phys}^2 = m^2 - \frac{\lambda m^2}{32\pi^2} \ln \frac{M^2}{m^2} \quad (9.12)$$

It looks puzzling that the physical mass depends on the arbitrary mass scale M that we introduced. But, this is remedied by the fact that the parameters m and λ depend on the mass scale M in such a way that m_{phys} is independent of M . Therefore, we must have,

$$\frac{dm_{phys}}{d \ln M} = 0 \quad (9.13)$$

This tells us,

$$0 = 2m \frac{dm}{d \ln M} - \frac{d\lambda}{d \ln M} \frac{m^2}{32\pi^2} \ln \frac{M^2}{m^2} - 2m \frac{dm}{d \ln M} \frac{\lambda}{32\pi^2} \ln \frac{M^2}{m^2} - \frac{\lambda m^2}{32\pi^2} \left(2 - \frac{2}{m} \frac{dm}{d \ln M} \right) \quad (9.14)$$

Since $m^2 = m_{phys}^2$ to zeroth order, we expect $\frac{dm}{d \ln M}$ to be atleast first order in λ . We will show in the next section that $\frac{d\lambda}{d \ln M}$ is second order in λ . Using these, we collect terms upto first order in λ to get,

$$2m \frac{dm}{d \ln M} = \frac{\lambda m^2}{16\pi^2} + \mathcal{O}(\lambda^2) \quad (9.15)$$

or,

$$\frac{dm}{d \ln M} = m \left(\frac{\lambda}{32\pi^2} + \mathcal{O}(\lambda^2) \right) =: m\gamma_m(\lambda) \quad (9.16)$$

$\gamma_m(\lambda)$ is called the *Anomalous mass dimensions*.

9.2 The *Beta Function* and Running of λ

In the previous chapter, we computed the amplitude for $\phi\phi \rightarrow \phi\phi$ process as,

$$\mathcal{A}(p_1, p_2 \rightarrow p_3, p_4) = \lambda + \lambda^2(V(s) + V(t) + V(u)) + \delta_\lambda \quad (9.17)$$

where,

$$V(p^2) = -\frac{1}{32\pi^2} \left[\left(\frac{2}{\epsilon} - \gamma \right) + \int_0^1 dx \ln \left\{ \frac{4\pi M^2}{m^2 + x(1-x)p^2} \right\} \right] \quad (9.18)$$

We previously choose to measure the scattering amplitude at $s = 4m^2$, $t = 0 = u$, and demanded that the scattering amplitude is the coupling constant λ itself. But now, we make another choice. We demand that $\mathcal{A} = \lambda_{phys} \neq \lambda$ for $s = 4m_{phys}^2$, $t = 0 = u$. But since $m_{phys}^2 = m^2 + \mathcal{O}(\lambda)$, we use $s = 4m^2$ from this stage itself. This gives us,

$$\lambda_{phys} = \lambda + \lambda^2 \left\{ -\frac{3}{32\pi^2} \left(\frac{2}{\epsilon} - \gamma \right) - \frac{1}{32\pi^2} \int_0^1 dx \left(\ln \left\{ \frac{4\pi M^2}{m^2 + 4x(1-x)m^2} \right\} + 2 \ln \left\{ \frac{4\pi M^2}{m^2} \right\} \right) \right\} + \delta_\lambda \quad (9.19)$$

Re-arranging things around, we get,

$$\lambda_{phys} = \lambda + \lambda^2 \left\{ -\frac{3}{32\pi^2} \left(\frac{2}{\epsilon} - \gamma \right) - \frac{3}{32\pi^2} \ln \left\{ \frac{4\pi M^2}{m^2} \right\} + \frac{1}{32\pi^2} \int_0^1 dx \ln \left\{ \frac{1}{1 + 4x(1-x)} \right\} \right\} + \delta_\lambda \quad (9.20)$$

We now choose the counter term to be,

$$\delta_\lambda = \lambda^2 \left[\frac{3}{32\pi^2} \left(\frac{2}{\epsilon} - \gamma \right) + \frac{3}{32\pi^2} \ln(4\pi) - \frac{1}{32\pi^2} \int_0^1 dx \ln \left\{ \frac{1}{1 + 4x(1-x)} \right\} \right] \quad (9.21)$$

This gives us,

$$\lambda_{phys} = \lambda - \frac{3\lambda^2}{32\pi^2} \ln \left\{ \frac{M^2}{m^2} \right\} \quad (9.22)$$

Now comes the key argument: λ_{phys} is the amplitude that we measured on an experiment. The measurement has nothing to do with the arbitrary scale M that we introduced previously. Hence, λ_{phys} *cannot* depend on M ! This tells us,

$$\frac{d\lambda_{phys}}{d \ln M} = 0 \quad (9.23)$$

This gives us,

$$0 = \frac{d\lambda}{d \ln M} - \frac{3\lambda}{16\pi^2} \frac{d\lambda}{d \ln M} \ln \left\{ \frac{M^2}{m^2} \right\} - \frac{3\lambda^2}{32\pi^2} \left(2 - \frac{2}{m} \frac{dm}{d \ln M} \right) \quad (9.24)$$

Now, we had $\frac{dm}{d \ln M} = \mathcal{O}(\lambda)$. We collect terms only to second order in λ , we get,

$$\frac{d\lambda}{d \ln M} \left(1 - \frac{3\lambda}{16\pi^2} \ln \left\{ \frac{M^2}{m^2} \right\} \right) = \frac{3\lambda^2}{16\pi^2} + \mathcal{O}(\lambda^3) \quad (9.25)$$

or,

$$\frac{d\lambda}{d \ln M} = \frac{3\lambda^2}{16\pi^2} + \mathcal{O}(\lambda^3) =: \beta(\lambda) \quad (9.26)$$

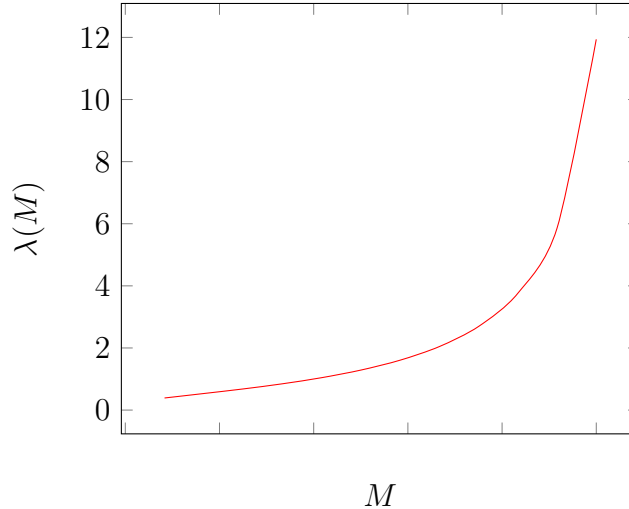
$\beta(\lambda)$ is called the *Beta function*. The beta function gives us how the coupling constant runs with the arbitrary scale M . To see this, let us solve the above differential equation. We have,

$$\int_{\lambda(M_0)}^{\lambda(M)} \frac{d\lambda}{\lambda^2} = \frac{3}{16\pi^2} \int_{\ln M_0}^{\ln M} d(\ln M) \quad (9.27)$$

This is easily solved and some re-arrangements later we get,

$$\lambda(M) = \frac{\lambda(M_0)}{1 - \frac{3\lambda(M_0)}{16\pi^2} \ln \frac{M}{M_0}} \quad (9.28)$$

To understand this, let us fix M_0 and vary M . Notice that since we are dividing by a number smaller than 1 when $M > M_0$, we have $\lambda(M) > \lambda(M_0)$ for $M > M_0$. The expression blows up when the denominator is zero. This gives us the plot,



We can see that the coupling constant diverges for large M . The point at which the coupling constant diverge is called the *Landau Pole*. There is an even pressing issue. The coupling constant becomes more than 1 after some value of M . We can no longer do perturbation theory for that range of values of M .

But what does all this mean? We might be inclined to think something along the lines of “the values of the coupling constants depends on the energy scale at which we probe at”. But this would be incorrect. Because M has nothing to do with the energy with which we probe the system. M is just an artifact of us doing dimensional regularization. Therefore, the differential equations for m and λ are just consistency conditions to ensure that the physically measured values do not depend on that arbitrary mass scale M ! There is nothing more to it. To see how coupling constants vary with energy scale of measurement, we must do the Wilsonian procedure of renormalization. (Read the next sentences with a pinch of salt) Then, it might be possible to draw a parallel between Wilson’s approach and the dim-reg approach and then it might be able to interpret the results we obtained in the “coupling constants as a function of energy scale” fashion.

Here is another possible way of interpreting the result: For a very high energy process with $s \gg m^2$ and $t = u = 0$, we have (after choosing the counter term),

$$\mathcal{A} = \lambda(M) - \# \lambda^2 \ln \frac{M^2}{s} \quad (9.29)$$

The amplitude $\mathcal{A}$ do not depend on M provided $\lambda(M)$ runs as the equation given by the beta function. Since we have the freedom to choose M , we choose it as $M^2 = s$. Then, we get the amplitude $\mathcal{A}$ as a function of s through the running equation. Then we can interpret the amplitude being dependent on the scale at which we probe the system.